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The Physiology and Pathophysiology of the Kidneys in Aging

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CHAPTER OUTLINE

GENERAL OVERVIEW OF THE BIOLOGY OF AGING

All, or nearly all, biological organisms exhibit the phenomenon of aging. The few exceptions are scientific curiosities.[1] Many speculations and hypotheses concerning the biological basis of aging have been advanced, but as yet none have provided a universal explanation for the aging process.[2] The life span of humans is regarded as limited at some maximum term that few approach.[1-3] Aging is an inevitable consequence of life. The rate of aging varies considerably, even among identical members of the same species,[1] an indication of the relatively minor role of heredity per se in determination of the life span of species. It has been estimated, from studies of monozygotic twins, that only 20% to 35% of the life span can be attributed to heredity (chromosomal or mitochondrial).[4] Epigenetics can also influence the biological clock.[5] Thus most theories of aging consider heredity, the environment, and chance to play variable roles in determination of the observed life span.[1] Metabolic rates and the balance of energy demand and supply appear to have a strong influence on life span.[6]

The fundamental biologic pathways accounting for diverse manifestations of aging have been the subject of intense investigation for many decades. Much progress has been made; however, many gaps in our knowledge of this complex process still remain,[7-32] including those of renal aging. At the cellular level, degeneration and faulty gene repair are core mechanisms of aging.[1] The net result is cellular and organ senescence. As indicated earlier, the rate of aging appears to be partially controlled by genetic pathways and biochemical processes. The explanation for the conservation of these processes that largely occur after maximal reproductive success is lacking.[1,3,4] As such, it is possible that many aspects of the aging process have not been evolutionarily conserved. Because of these factors, it seems likely that the biological events responsible for aging may differ among species, complicating the study of aging enormously—studies of aging in experimental animals may therefore have limited relevance to human aging.

At the organ and cellular level, the major hallmarks of aging are genomic instability, epigenetic alterations, mitochondrial dysfunction, dysregulated nutrient sensing, telomere attrition, loss of protein homeostasis, stem cell exhaustion, accumulation of senescent cells, oxidation and glycation of tissue proteins, and altered intercellular communication (reviewed by Sturmlechner and coworkers[28] and López-Otín and associates).[33,34]

A detailed description of these processes underlying aging is beyond the scope of this brief introductory review. Age-dependent accumulation of stochastic damage to critical molecular pathways seems to be a dominant driver of aging.[1] The generation of ATP to provide energy to sustain cellular function is key for life, and this process is systematically altered in aging.[35]

Energy production, via mitochondrial dysfunction, regularly declines with age,[1,35] possibly due to stochastic

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damage to mitochondrial DNA and inefficient repair. Oxidative damage to DNA has been postulated to be one of the root causes of aging.[6,36] Mitochondrial inefficiency can lead to impaired cellular autophagy and cellular senescence.[37-40] Alternate hypotheses of defects in oxidative metabolism leading to a shift from aerobic to anaerobic metabolism are also possible.[6] A mismatch occurs in aging between lowered energy demand and excess supply. Caloric restriction has been found to be one of the most powerful experimental means of retarding the aging process.[33,41] Sirtuins, nicotinamide adenine dinucleotide (NAD^+)-dependent proteins, which deacetylate crucial enzymes in the oxidative metabolic pathway, may be responsible for this effect.[11,12,33,42,43] Sirtuin production diminishes with age and promotion of sirtuin production, as by the administration of mammalian target of rapamycin (mTOR) inhibitors, which mimic caloric restriction, can prolong life span.[33] This is known as the "bioenergetic theory of aging."[2] Whether a genetic program (the aging clock) defines the decline in bioenergetics with aging is unclear.[44] Epigenetic alterations and DNA methylation may account for some of these variations.[45]

Aging is also associated with shortening of the telomeres in nuclear DNA due to changes in the activity of telomerase, an enzyme essential for maintaining telomere length in dividing cells.[2,46,47] Gradual loss of telomeres leads to cessation of cell division after a maximum number of cell divisions, known as the "Hayflick limit," and induces senescence, ultimately leading to cell death.[48]

The individual organ systems of the human body exhibit phenomena connected to the overall aging process, also at variable rates. These organ-based manifestations of aging include loss of skin elasticity, decreases in hair pigmentation, slowing of nerve impulses, decreased mineral density of bone, decreased compliance of major vessels, decreased forced expiratory lung volume, decreased muscle mass, and reduced metabolic rate. The kidneys share in these inevitable biologic consequences of aging, as we will detail here. Numerous reviews and treatises covering the general topic of renal aging have been published over the past 4 decades.[7-32] It seems likely that the complex processes operating in aging at the organism level also play important roles in the manifestations of organ-based senescence, such as that displayed in the kidneys.

The genetic component of renal aging has been analyzed by genome-wide association studies and transcriptomics,[49-51] and several candidate loci and factors have been identified. Increased oxidative or glycative stress,[52,53] dysregulation of autophagy,[54] reduced Klotho generation,[16,55-58] enhanced fibrosis,[59,60] increased capillary rarefaction,[10] and increased activation of the angiotensin II type 1 receptor[61-63] may all contribute to renal aging. The alpha-Klotho protein is regarded as critical to the aging process, as a natural inhibitor of aging. The alpha-Klotho protein functions as a receptor for fibroblast growth factor 23 and thereby regulated phosphate and vitamin D metabolism, particularly in the kidney. Modulation of Klotho activity may modify rates of organ aging and senescence.[64] Klotho deficiency also promotes inflammation.[55,56] Toxins such as D-serine[65] might be involved in the development of fibrosis in aging. Experimental evidence (in mice) has suggested that the level of fructokinase activity may be essential for renal aging.[66] In addition, endostatin and transglutaminase activity may contribute to the renal fibrosis of aging according to studies in mice, but these effects might be species specific.[67] Factors postulated to be involved in renal aging are summarized in Box 21.1.

HEALTHY AGING AND AGE-RELATED COMORBIDITY

The normal aging process leads to subtle and cumulative alterations in cellular and organ function and also predisposes an individual to certain age-related conditions, including atherosclerosis, hypertension, type 2 diabetes, cancer, osteoporosis, and dementia. Disentanglement of these disease states from phenomena that might be called "healthy" aging can be challenging. Healthy aging might be defined as the state that is universal, or nearly so, in all aging subjects, whereas age-related comorbidities affect only some of the aging population and involve processes that are disease specific. For example, a decline in bone density or forced expiratory lung volume is characteristic of aging; defining a threshold for a disease-associated decline in these functions requires comparison with the expected changes with healthy aging. Type 2 diabetes prevalence increases with aging, and the aging process leads to beta cell exhaustion in the pancreatic islet cells. Therefore aging per se predisposes some older subjects to develop overt diabetes.[68] The best examples of healthy aging are individuals selected for the donation of one kidney for renal transplantation, but even this "healthiest of the healthy" may not be entirely normal because they also may be mildly obese, have treated mild hypertension, or have covert disorders not easily identifiable, such as low nephron endowment stemming from in-utero impaired nephrogenesis, even after an exhaustive

Box 21.1 Some factors postulated to be involved in renal aging

↓Sirtuin 1/6 (a histone deacetylase enzyme)
↓Klotho expression (and Wnt signaling)
↓Antioxidant production,↑oxidant activity
↓ Energy demand: ↑energy supply (mitochondrial dysfunction)
↑Telomere shortening
↑ DNA damage repair
↑ DNA methylation
↑Angiotensin II receptor signaling (via Wnt)
↑Cell cycle arrest (GI, via P16ink)
↓Autophagy
↑Fibrosis (transforming growth factor beta [TGF-β]–mediated)
↓Elimination of senescent cells
↓Insulin-like growth factor 1 (IGF-1) signaling
↓Proliferator-activated receptor-γ (PPARγ) activity
↑Capillary rarefaction
↑Podocyte apoptosis or detachment (podocytopenia—absolute and/or relative to capillary surface area)
↑Vascular sclerosis and glomerular ischemia
↑Advanced glycation end products
↑D-Serine toxicity
↑ Endostatin, transglutaminase activation

GI, Gastrointestinal.

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clinical, laboratory, and imaging-based pretransplantation evaluation. Studies of older, apparently healthy, adults in the community are inevitably an admixture of healthy aging and aging with comorbidity, sometimes clinically unrecognized.[69] In this context, analyses of low-molecular-weight proteins in urine obtained from supposedly healthy subjects have suggested resemblances between healthy aging and chronic kidney disease (CKD), but these findings might be the consequence of overlapping comorbidities (covert and overt) with aging per se[70] or an acceleration of normal aging by the effects of CKD. In this chapter, we will use the adult living kidney donor as the prototype for healthy aging, recognizing some of the pitfalls in this assumption. A detailed discussion on the approach to management of kidney disease in older adults can be found in Chapter 60.

DIFFERENCES BETWEEN HUMANS AND OTHER ANIMALS

Although many of the fundamental cellular processes of biologic aging are evolutionarily conserved, disparities at the organ-system level may exist between species concerning the anatomic and functional consequences of aging. For example, many animal species (e.g., murine species) continue to grow throughout their life span, whereas humans cease growing after attaining maturity. Additionally, nonhuman species adapted to confinement and regular feeding rather than foraging may have different aging processes in play. The growth regulatory genetic program might be evolutionarily conserved among mammals, but the pace of growth is modulated in larger animals including humans.[71] Metabolic demand may therefore differ among aging humans and aging experimental animals. Such differences can have profound effects on the organ-based manifestations of aging including the kidneys. For example, arteriosclerosis is a defining feature of human kidney aging not seen in rodents, whereas focal segmental glomerulosclerosis is a feature of rodent aging not seen in humans from aging alone. Thus one needs to be cautious about inferring mechanisms of organ aging in humans from studies of experimental animals or lower organisms. The circumstances of gestation and birth, unique to humans, and in utero organogenesis can also have effects later in life.[72]

ANATOMIC CHANGES OF THE KIDNEYS WITH AGING

KIDNEY SIZE AND VOLUME

Gourtsoyiannis and colleagues[73] analyzed computed tomography (CT) scans from 360 patients with no kidney disease to estimate kidney parenchymal thickness. They found that for each decade of older age, kidney parenchymal thickness decreased about 10%. Another CT study in 1040 asymptomatic adults found that the factors associated with larger kidney size were male sex, taller height, and larger body mass index, whereas older age and renal artery stenosis were associated with smaller kidney size.[74] Another study of 1056 patients showed similar findings and provided evidence that atherosclerosis accelerated the decline in kidney size with advancing age.[75] In a population-wide study of kidney size by ultrasound in Sardinia in healthy subjects aged 18 to 100 years, Piras and coworkers[76] found that kidney size diminished starting in the fourth to fifth decades in men but decreased over the entire adult span in women.

A large magnetic resonance imaging study assessed total kidney volume in 1852 subjects from the Framingham Heart Study.[77] The authors reported that the average decline in total kidney volume in both sexes was 16.3 cm^3/decade, with a more pronounced decline beyond age 60 and a more pronounced decline in men than women. In addition, using a subset of 196 apparently healthy women and 112 apparently healthy men, this study determined the sex-specific upper and lower 10th percentile thresholds for total kidney volume. In multivariable models, kidney volume above the 90th percentile was associated with younger age (odds ratio, 0.67), whereas kidney volume below the 10th percentile was associated with older age (odds ratio, 1.67).

KIDNEY CORTEX AND MEDULLA

Wang and colleagues[78] have studied the CT scans of 1344 potential living kidney donors up to 75 years of age. They found a stable kidney parenchymal volume in those up to 50 years of age and a decline thereafter.[78] This study also obtained the volumes of the kidney cortex (average, 73% of parenchymal volume) and medulla (average, 27% of parenchymal volume) separately. An increasing medullary volume with age seems to attenuate some of the loss of total kidney parenchymal volume with age due to decreasing cortical volume (Fig. 21.1). Besides increased medullary volume, other studies have suggested that an increase in renal sinus fat with older age also masks some of the observed decline in kidney volume.[73,79] A subsequent study in 2876 kidney donors characterized the number and size of kidney medullary pyramids with age.[80] With older age there was an increase in the size but decrease in the number of medullary

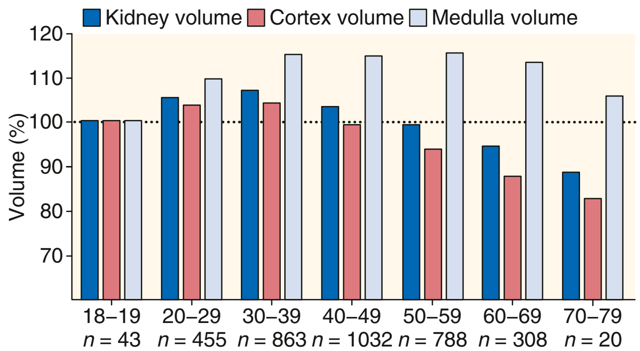

Fig. 21.1 Total kidney, cortical, and medullary volumes among 3509 potential kidney donors in the Aging Kidney Anatomy study (expansion of previously reported findings). Among all donors, older than 40 years, cortical volume decreases and medullary volume increases, making total kidney volume relatively constant until 50 years of age. Beyond that, medullary volume no longer increases, leading to a decrease in total kidney volume from the decreasing cortical volume. Findings were compared proportional to the respective volumes in the youngest age group (18–19 years). (From Wang X, Vrtiska TJ, Avula RT, et al. Age, kidney function, and risk factors associate differently with cortical and medullary volumes of the kidney. *Kidney Int.* 2014;85:677–685.)

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pyramids. Findings were consistent with medullary pyramids merging with age due to loss of detectable cortical columns separating pyramids from age-related nephrosclerosis.

KIDNEY CYSTS

Kidney cysts are relatively common, of tubular diverticuli origin,[81] and their frequency and size increase with older age.[82,83] There has been increased detection of cysts due to technologic advancements with imaging modalities, so it is important to consider cyst size thresholds when counting cysts.[82] A study of 1948 potential kidney donors demonstrated that even in this predominantly healthy population, cortical and medullary cysts larger than or equal to 5 mm were more frequent in older men and were associated with larger body surface area, albuminuria, and hypertension.[82] Moreover, this study generated upper reference limits (97.5th percentile) for the number of cysts in both sexes by age. The upper reference limit for the number of cysts in both kidneys of 18- to 29-year-olds is one for both men and women but increases to 10 in men and 4 in women older than 60 years. Besides simple kidney cysts, parapelvic cysts, hyperdense cysts, angiomyolipomas, and cysts or tumors suspicious for cancer are also more frequent with older age.[82] Parapelvic cysts are thought to have a lymphatic origin[84] and increase with age, but are not associated with hypertension or albuminuria.

OTHER STRUCTURAL CHANGES

Other kidney parenchymal changes that become more prevalent with older age in ostensibly healthy adults include calcifications, focal cortical scars, fibromuscular dysplasia, and renal artery atherosclerosis without stenosis.[85] Of these, the two findings most strongly associated with age are atherosclerosis of renal arteries and focal cortical scarring. In donors younger than 30 years, the prevalence of atherosclerosis and focal scarring were 0.4% and 1.5%, respectively. However, in donors older than 60 years, the prevalence of atherosclerosis was nearly 25%, and the prevalence of focal scarring was 8%. Focal scarring also contributes to an increase in kidney surface roughness with age.[86]

MICROSCOPIC CHANGES

GLOMERULI

Number of glomeruli

Several different approaches have been used to count the total number of glomeruli per kidney, with similar findings. Autopsy studies have shown that on average, humans have 900,000 glomeruli/kidney, but with significant variability, ranging from as low as 200,000 up to 2.7 million glomeruli/kidney.[87,88] Lower glomerular number counts are likely the result of both the reduction in nephrogenesis stemming from intrauterine events (low nephron endowment) and the normal progressive loss of glomeruli with age. Counts of the glomerular number have been obtained from autopsy studies, where complete sectioning of the kidney can be performed.[89] However, many autopsy studies have not distinguished normal nonsclerotic glomeruli from globally sclerotic glomeruli in their counts. Furthermore, the comorbid state in autopsy studies (often unknown, with sudden unexpected deaths) may increase nephron loss beyond that expected from aging alone.

Living kidney donors who were carefully screened to confirm health provide a unique opportunity to determine how the glomerular number relates to aging. A first large study in 1638 living kidney donors estimated numbers of nonsclerotic and globally sclerotic glomeruli using the predonation CT and implantation biopsy.[88] A subsequent study of 3020 donors improved the estimated nephron number after excluding patients with family history of end-stage kidney disease (ESKD) and correcting for several biases, such as accounting for missing glomerular tufts, correcting for thicker CT axial image sections, and using a more accurate shrinkage factor due to formalin fixation.[90] This new method led to a nephron number that is approximately 27% higher than prior estimates. The youngest male and female kidney donors (18–39 years of age) had a mean number of nonsclerotic glomeruli of about 1.4 million and 1.25 million per kidney, respectively. The oldest male and female kidney donors (70–77 years) had about 1.07 million and 0.83 million nonsclerotic glomeruli per kidney, translating to about 24% nephron loss in males and 33% in females. The youngest male and female kidney donors had 1.3% and 1.5% globally sclerotic glomeruli, respectively. The percentage of globally sclerotic glomeruli in the oldest male and female kidney donors was 12.6%, translating to an approximately 11% increase. Thus the 24-33% loss in nonsclerotic glomeruli, but with only an 11% increase in globally sclerotic glomeruli, supports the hypothesis that globally sclerotic glomeruli are eventually completely reabsorbed or significantly atrophied so that they can no longer be detected on tissue sections examined by light microscopy (Fig. 21.2). This is in agreement with an older study by Hayman and associates, who postulated that "scars of destroyed glomeruli disappear without leaving recognizable traces."[91] The concept of missing or reabsorbed glomeruli is very important because a standard pathology report of the percentage of glomerulosclerosis on a kidney biopsy may substantially underappreciate the true age-related loss of glomeruli.

> ***Clinical Relevance***
> The reabsorption of globally sclerosed glomeruli with age may lead to an underappreciation of older kidneys according to kidney biopsy assessment.

Glomerulosclerosis

An increasing percentage of global glomerulosclerosis is a feature of an aging kidney that has been demonstrated in autopsy and living kidney donor studies.[89,92,93] An autopsy study of 58 kidneys (48 from adults) from a mixed population of Australian Aborigines and non-Aborigines, U.S. Caucasians, and African-Americans has shown a range of 0% to 23% with global glomerulosclerosis and a strong association with older age.[89] A study of 1203 living kidney donors has confirmed these findings and showed an increasing prevalence of focal and global glomerulosclerosis but not focal and segmental glomerulosclerosis (FSGS) with older age.[92] Whereas the prevalence of any global glomerulosclerosis was only 19% in the youngest kidney

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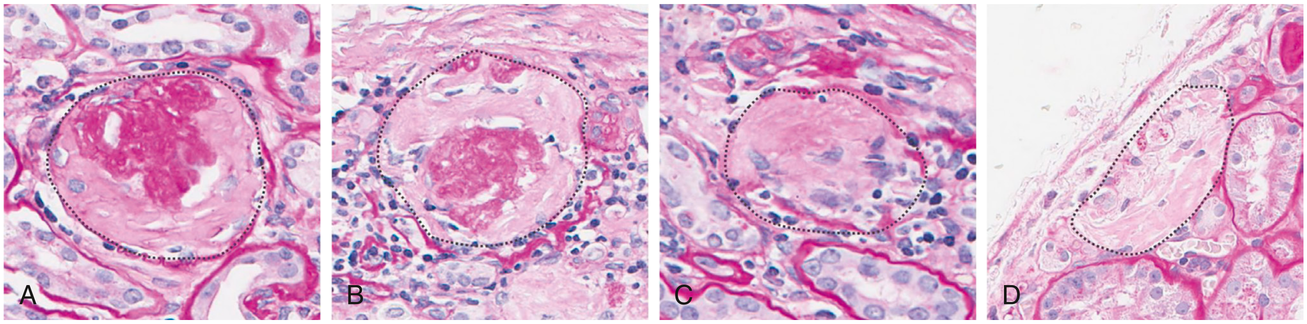

Fig. 21.2 Examples of globally sclerosed glomeruli (GSG) in different stages of their involution. (A and, B) GSG are easily discernible in the biopsy sample. (C and D) GSG that may be overlooked because they progressively atrophy and lose a discernible capsule. All GSG are traced with a *black dashed line.*

Table 21.1 Expected number of globally sclerotic glomeruli per biopsy section[a]

Age group, years	Number of glomeruli group (mean)[b]					
	1.0-4.0(4.0)	4.1-8.0(6.3)	8.1-16.0 (12.2)	17.1-32.0(22.8)	32.1-48.0(37.8)	48.1-91.4(59.8)
18-29	0.5	0.5	1.0	1.0	1.0	1.5
30-39	0.5	1.0	1.0	1.5	2.0	2.5
40-49	0.5	1.0	1.5	2.0	2.5	3.5
50-59	1.0	1.5	1.5	3.0	4.0	5.0
60-69	1.0	2.0	2.5	4.0	5.5	6.5
70-77	1.5	2.5	4.0	5.5	7.0	8.5

[a]As predicted by the total number of glomeruli and age from the Aging Kidney Anatomy study.
[b]Number of glomeruli was sum of all complete glomerular profiles (including empty capsules) + .675 × (partial glomerular profiles bisected by the needle).
Adapted from Asghar MS, Denic A, Mullan AF, et al. Age-based versus young-adult thresholds for nephrosclerosis on kidney biopsy and prognostic implications for chronic kidney disease. *J Am Soc Nephrol*. 2023;34(8):1421–1432.

donors (age 18–29 years), the prevalence was 82% among the 11 oldest donors (70–77 years of age). Recently in 2583 living kidney donors who were normotensive across three centers, the upper 95th percentile reference limit of glomerulosclerosis expected for age was determined (Table 21.1). For example, for a 25-year-old who has a biopsy with 16 glomeruli, up to one globally sclerotic glomerulus on biopsy might be expected before suspecting glomerulosclerosis from disease. However, for a 76-year-old who has a biopsy with 16 glomeruli, up to 4 globally sclerotic glomeruli on biopsy might be expected before suspecting glomerulosclerosis from disease. Using kidney donor biopsies with 48 to 91 glomeruli on a biopsy, the absolute percentage of 95th percentile threshold can be calculated.[94] For example, for the youngest kidney donors (18–29 years of age), the 95th percentile expected for age is 1.7%, and this increases to 16% for the oldest donors (70–76 years of age) (Fig. 21.3). Among the 5% of donors in whom the number of globally sclerotic glomeruli was higher than the 95th percentile expected for age, there was a higher prevalence of hypertension and interstitial fibrosis and a higher percentage of ischemic-appearing glomeruli (pericapsular fibrosis, capsular thickening, and capillary loop wrinkling).[95] The use of these age-specific reference limits for globally sclerotic glomeruli can help to predict which patients will develop progressive CKD.[96]

Certain morphologic findings of globally sclerotic glomeruli should also be taken into consideration; for example, solidification forms of glomerulosclerosis are always pathologic and are never simply an age-related finding.[97-99]

Clinical Relevance
Glomerulosclerosis associated with aging and disease can both be global and obsolescent in appearance. In contrast segmental and solidification forms of glomerulosclerosis, only occur with disease.

The obsolescent type of morphology of global glomerulosclerosis is characterized by the filling of the Bowman space with collagenous material, accompanied by retraction of the capillary tuft. This type of global glomerulosclerosis is observed with normal aging or in with chronic kidney disease. The obsolescent-type global glomerulosclerosis is presumed to reflect an ischemic process. FSGS is not associated with healthy aging in humans, unlike murine species. The lesion of FSGS in growing rats is likely to be mediated by podocyte injury and detachment. FSGS lesions in humans almost always indicate an underlying pathologic disease process and not only senescence.[97-99] The possession of two risk alleles for APOL1 appears to enhance age-related global glomerulosclerosis in individuals of West African ancestry.[100,101] This phenomenon might help explain in part the higher risk in Black individuals for the development of hypertension and glomerular filtration rate (GFR) decline at an earlier age.[102]

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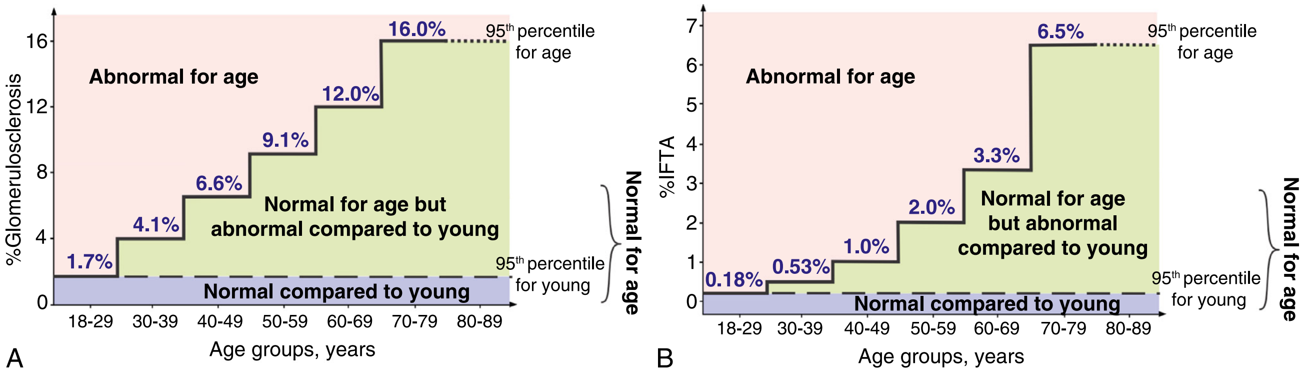

Fig. 21.3 Thresholds for identifying abnormal levels of nephrosclerosis were based on the 95th percentile for 2583 normotensive living kidney donors. (A and B) There were no normotensive kidney donors older than the age of 77 years, so the *dotted line* in the 80 to 89 years age group was extended using thresholds in donors ages 70 to 77 years. For the nephrosclerosis measures of %Glomerulosclerosis, and % interstitial fibrosis and tubular atrophy (IFTA), the "abnormal compared to young" threshold was 1.7%, and 0.18%, whereas the "abnormal for age" threshold ranged from 1.7% to 16% and 0.18% to 6.5%, respectively. (Modified from Asghar MS, Denic A, Mulan AF, et al. *J Am Soc Nephrol.* 2023;23[8]:1421–1432.)

Glomerular hypertrophy

Entire nephrons hypertrophy in diabetes and obesity,[103-106] conditions that have become more prevalent with older age in developed and developing countries. Nephron hypertrophy is also seen in subjects with low nephron endowment at birth.[107] Although hypertrophy occurs in the glomerular and tubular segments of a nephron unit,[108] glomerulomegaly is much easier to appreciate on visual inspection of a kidney biopsy than tubular enlargement. The relationship between glomerular size and older age needs to account for the underlying study population (e.g., age-related comorbidities such as diabetes and obesity or nephron endowment at birth) and whether glomerular size was estimated from only nonsclerotic glomeruli or nonsclerosed and the smaller globally sclerosed glomeruli (GSG) that become more common with advancing age.

Published studies have reported conflicting results, likely related to these issues, with some showing an increase in glomerular size with age[109-111] and others showing a decrease in glomerular size with age.[112,113] Studies that were limited to carefully screened healthy living kidney donors and only studied nonsclerosed glomeruli have found no change in glomerular size with aging.[86,88,114] Nonsclerosed glomerular volume in patients with kidney tumors who underwent nephrectomy was stable until 75 years of age, similar to findings in living kidney donors.[115] However, beyond 75 years of age, nonsclerosed glomerular volume decreased due to a higher proportion of smaller, ischemic-appearing glomeruli. Given this lack of increase in glomerular volume, albuminuria would not be expected with aging alone in humans because albuminuria occurs with glomerular hypertrophy via a disorganized glomerular structure unable to efficiently prevent protein leaking.[111] Increased glomerular volume (glomerular hypertrophy) and glomerulosclerosis are associated with comorbidities common in older persons, including obesity, type 2 diabetes, hypertension, and proteinuria. Such hypertrophy is also associated with male sex, taller height, and a family history of ESKD.[106,114]

TUBULES

Tubular hypertrophy

Notably, glomeruli only comprise about 4% of the total cortical volume.[88] The remaining parenchymal volume is largely composed of renal tubules. Although glomerular enlargement does not occur with healthy aging, there is some evidence of tubular enlargement with age. A study of 1367 patients with kidney tumors assessed proximal and distal tubular diameters and found that an age-related increase in tubular size occurred in distal tubules at all depths but not proximal tubules.[116] Increase in distal tubular size, as well as atrophy and disappearance of globally sclerotic glomeruli, disperses the remaining glomeruli farther apart from each other, thereby decreasing glomerular density with aging.[117] However, the relationship between glomerular (nonsclerosed and globally sclerosed) density varies, depending on how much global glomerulosclerosis is present in the section biopsied. In biopsy sections with less than 10% global glomerulosclerosis, glomerular density is diminished with older age, whereas in biopsy sections with more than 10% global glomerulosclerosis, glomerular density is increased with older age.[117] In regions of the renal cortex without significant glomerulosclerosis, age-related distal tubular enlargement disperses glomeruli farther apart, decreasing their density. However, in regions with significant glomerulosclerosis, the overall atrophy of glomeruli and their tubules brings the glomeruli closer together, thereby increasing their density (Fig. 21.4).

Tubular diverticuli

Darmady and colleagues[111] found in an autopsy study that tubular diverticuli increase with older age. This finding parallels two other age-related findings in healthy adults. First, simple parenchymal cysts, which likely originate from these diverticuli,[81] become more frequent with older age. Second, the mean profile tubular area increases with older age.[86] Enlargement of tubules, with upregulation of growth factors, may contribute to the development of diverticuli, with the eventual formation of cysts.[118,119]

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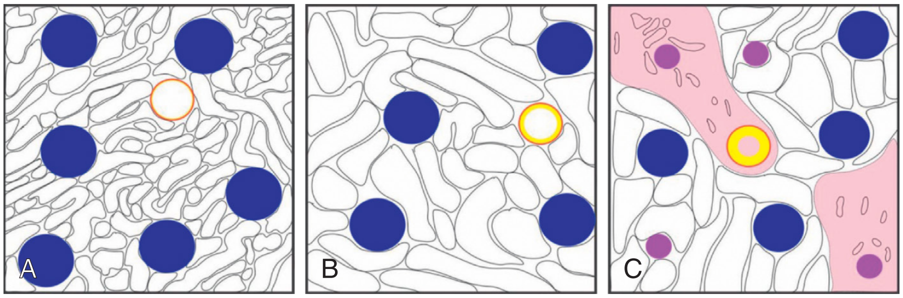

Fig. 21.4 Schematic illustration of how the percentage of glomerulosclerosis influences glomerular density. (A) Example of a young individual with a certain glomerular density *(blue solid circle),* normal tubules, and normal artery *(orange circle),* with minimal to no intimal thickening *(yellow circle).* (B) In an older person in whom the biopsy shows less than 10% glomerulosclerosis, tubules become larger and disperse the glomeruli apart and their density decreases. (C) In older persons in whom there is more than 10% glomerulosclerosis *(pink solid circles)*, the associated interstitial fibrosis and tubular atrophy *(light pink areas)* bring the glomeruli closer together, thus increasing their density. Arteriosclerosis with intimal thickening *(thicker yellow line)* is also pronounced in these regions in older adults.

ARTERIES AND ARTERIOLES

Arteriosclerosis and arteriolar hyalinosis are two common findings in the kidney biopsy of normal adults that become more prevalent with age.[86] Arteriosclerosis (luminal stenosis by intimal thickening) with aging may cause down-stream ischemic injury, leading to a lower cortical-to-medullary volume ratio, even in healthy adults.[86] Arteriolar hyalinosis increases with age and is associated with renal artery atherosclerosis in healthy adults and hypertensive patients.[120-122] Older age and hypertension may lead to global glomerulosclerosis and nephrosclerosis via reduced blood flow due to ischemic injury from the narrowing of small arteries and arterioles.[123] Notably, lower total nephron number per kidney is associated with arteriosclerosis, even after adjusting for age and sex.[88]

NEPHROSCLEROSIS AND INTERSTITIAL FIBROSIS

The term *nephrosclerosis* describes a microstructural biopsy pattern of global glomerulosclerosis, arteriosclerosis, and interstitial fibrosis with tubular atrophy. Nephrosclerosis is notably seen with hypertension[124] but is also described in healthy older kidney donors without hypertension or only mild hypertension.[92] Thus the term "hypertensive nephrosclerosis" may be a misnomer. Nephrosclerosis is thought to be due to increased intimal thickening of small arteries in the kidneys (arteriosclerosis, arteriolosclerosis), leading to glomerular ischemia, with the wrinkling of capillary tufts, thickening of the basement membrane, and progressive pericapsular fibrosis.[124,125] Simultaneously, proteinaceous material accumulates in the Bowman space, likely due to a combination of a disturbed balance between formation and breakdown of the glomerular extracellular matrix[126] and podocyte depletion,[92,110,126] probably due to perturbations in local mechanical forces.[127] Eventually, ischemic glomerular tufts fully collapse, forming GSG. As a consequence of glomerulosclerosis, accompanying tubules atrophy with an accumulation of surrounding interstitial fibrosis.

All four features of nephrosclerosis—global glomerulosclerosis (GSG), arteriosclerosis, IFTA—increase with older age and, when combined together (at equal weighting) they represent a sclerosis score (Fig. 21.5).[92] If we define nephrosclerosis as the presence of at least two of the previously mentioned abnormalities, then from Fig. 21.5 we can see that the youngest donors (18–19 years of age) do not have detectable nephrosclerosis. The prevalence of nephrosclerosis in 20- to 29-year-old donors is 5% and rises to 69% among the oldest donors.

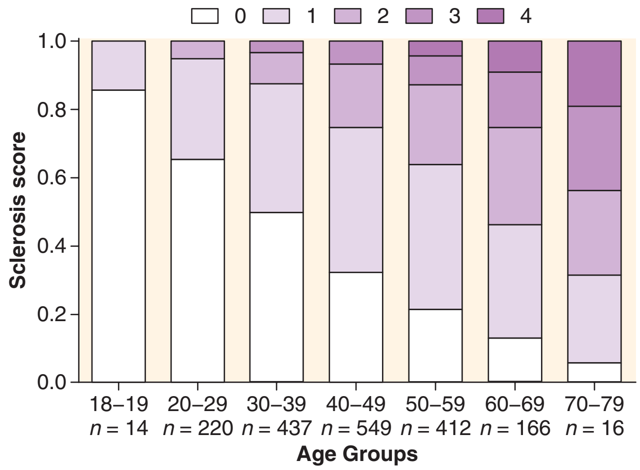

Fig. 21.5 The nephrosclerosis score increases with age among 1814 living kidney donors from the Aging Kidney Anatomy study (expansion of previously reported findings). The total nephrosclerosis score is obtained by adding the individual scores of histologic abnormalities: any global glomerulosclerosis, any arteriosclerosis, interstitial fibrosis greater than 5%, and any tubular atrophy. *White bars* in all age groups represent a score of 0 (no abnormalities present), and a *dark purple bar* represents a score of 4 (presence of all four pathologic abnormalities). Three intermediate scores are presented with different shades of *purple color*. (From Rule AD, Amer H, Cornell LD, et al. The association between age and nephrosclerosis on renal biopsy among healthy adults. *Ann Intern Med.* 2010;152:561–567.)

The number and size of the nephrons are the primary determinants of the kidney cortical volume. A combination of age-related glomerulosclerosis with corresponding IFTA is responsible for the observed loss of cortical volume with healthy aging seen in living kidney donors. This nephrosclerosis typically starts in glomeruli in the superficial cortical zones, as evident by the high proportion of global glomerulosclerosis in this region compared with middle and deep regions in older adults.[128] Because the corresponding tubules of these superficial glomeruli contribute more to the cortical volume, their atrophy with age-related nephrosclerosis leads to a decrease in cortical volume with aging. There is less evidence of GSG with age

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in the deeper cortex among nephrons that descend into the medulla. Despite a 24% to 33% decrease in nephron number from ages 18 to 76 years, there is only a 17% decrease in cortical volume and only a 12% decrease in total kidney volume due to hypertrophy of the distal tubule.[73,79]

One might expect that with age-related nephrosclerosis, there would be compensatory hypertrophy of the remaining, functional glomeruli. However, studies in living kidney donors have not confirmed this hypothesis.[117] A possible explanation may be that with healthy human aging, there is a decreased metabolic demand for glomerular function, so the loss of glomeruli does not cause the enlargement of the remaining glomeruli. Notably, comorbidities that can become more prevalent in older age, such as obesity and type 2 diabetes with albuminuria, associate with glomerular hypertrophy.[117] Also, low nephron endowment at birth or disorders that accelerate nephron loss with advancing age can contribute to glomerulomegaly observed in adulthood.

Similar to age-based thresholds in glomerulosclerosis, the upper 95th percentile reference limit of IFTA has been described.[94] For example, for the youngest kidney donors (18–29 years of age), the 95th percentile expected for age was 0.18% IFTA, which increased to 6.5% for the oldest donors (70–76 years of age) (Fig. 21.3).

FUNCTIONAL CHANGES OF THE KIDNEYS WITH AGING

These include changes in the GFR and other functional changes.

GLOMERULAR FILTRATION RATE

WHOLE-KIDNEY GLOMERULAR FILTRATION RATE

In a series of pioneering studies begun more than 60 years ago, Davies and Shock examined 70 healthy adult men between the ages of 24 and 89 years and, in a cross-sectional analysis, showed (by urinary inulin clearance) a monotonically lower GFR in subjects older than 30 years.[129] The GFR in the oldest group was found to be 46% less as compared with the GFR in the youngest group. Several decades later, Lindeman and colleagues[130] carried out a longitudinal study (the first of its kind) by following 254 mostly healthy adult individuals, although some had diabetes, for up to 14 years. They found that the mean annual decline in GFR, estimated by 24-hour urinary creatinine clearance, was 7.5 mL/min per decade; however, one third of the subjects had no decrease in kidney function and a small subset experienced an increase in GFR. In addition to measurement error explaining this finding, a rise in GFR with age might also be explained by onset of comorbidities (e.g., obesity, type 2 diabetes) that cause hyperfiltration and become more common with aging.[131] This longitudinal rate of GFR decline is similar to the cross-sectional decline of 6.3 mL/min per decade obtained from a study of potential kidney donors (GFR measured by iothalamate clearance).[92] Among 4500 potential kidney donors in the expanded analysis from the Aging Kidney Anatomy study (Fig. 21.6), the average decline per decade for estimated

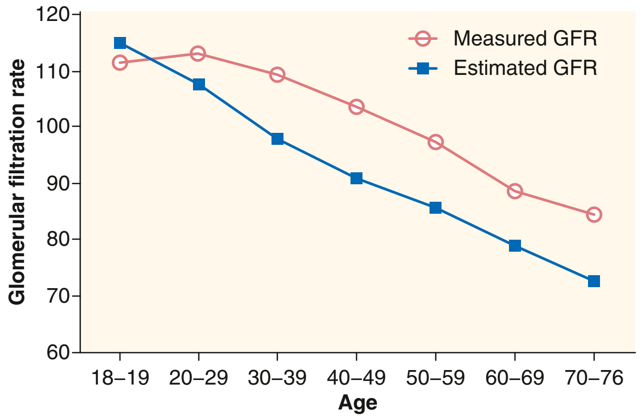

Fig. 21.6 Age-related glomerular filtration rate (*GFR;* mean values) decline among 4500 potential kidney donors in the Aging Kidney Anatomy study (expansion of previously reported findings). Decline in the estimated GFR (CKD-EPI study equation; see text) is calculated to be 7.4 mL/min/decade and 6.1 mL/min/decade for the measured GFR. (From Rule AD, Amer H, Cornell LD, et al. The association between age and nephrosclerosis on renal biopsy among healthy adults. *Ann Intern Med.* 2010;152:561–567.)

GFR was 7.4 mL/min/1.73 m^2, whereas for measured GFR (iothalamate clearance) it was 6.1 mL/min/1.73 m^2.

Although Figure 21.6 shows mean estimated GFR (eGFR) and measured GFR, there is variability within each age group (Table 21.2).[132] It can be appreciated that both upper and lower reference limits also decline with aging. Similar findings in healthy donors have been reported by Pottel and colleagues.[133] Heathy aging is associated with a higher measured GFR compared with unhealthy aging (with comorbidity).[134] Other longitudinal studies of the decline in kidney function (GFR or creatinine clearance) with aging have shown generally similar findings to those of Lindeman and associates,[130] including the relative stability of kidney function for extended periods in some persons who were otherwise experiencing healthy aging.[135-137] In addition, in these studies, the rate of decline of kidney function appeared to increase with each passing decade in persons beyond the fourth decade.

Among patients who underwent radical or partial nephrectomy for a kidney tumor and who had a minimal amount of nephrosclerosis on biopsy, there was still a substantial decrease in eGFR of about 35 mL/min/1.73 m^2 from age 25 years to 75 years.[138,139] Notably, this decline in eGFR with 5 decades of aging among persons with minimal nephrosclerosis is about double the decrease in eGFR from minimal to moderate nephrosclerosis at the same age (17 mL/min/1.73 m^2).

SINGLE-NEPHRON GLOMERULAR FILTRATION RATE

Several studies have found that a higher whole-kidney GFR correlates with larger kidneys or larger kidney cortical volume.[77,78,86] The age-related nephron loss due to increased nephrosclerosis is followed by a parallel whole-kidney GFR decline.[88] There is no compensatory increase in single-nephron GFR (snGFR) as nephrons are lost with aging.

Although in vivo measurements of snGFR through glomerular micropuncture are possible and have been done in animal studies,[140,141] a direct measure of snGFR (by micropuncture) is not feasible or safe in humans. However,

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Table 21.2 Reference values for estimated and measured glomerular filtration rate in potential kidney donors[a]

Age group (years)	No.	Estimated glomerular filtration rate (GFR) (measured GFR)			
		Fifth percentile	Median	Mean	95th percentile
18–19	46	88 (87)	114 (107)	115 (111)	145 (144)
20–29	584	81 (84)	108 (111)	108 (113)	130 (149)
30–39	1090	75 (82)	98 (106)	98 (109)	118 (147)
40–49	1364	69 (77)	91 (102)	91 (104)	110 (135)
50–59	1001	65 (72)	85 (94)	85 (97)	103 (133)
60–69	379	61 (64)	78 (87)	79 (88)	97 (118)
70–76	36	55 (57)	71 (85)	73 (84)	91 (113)

[a]Expansion of previously reported findings.
From Murata K, Baumann NA, Saenger AK, et al. Relative performance of the MDRD and CKD-EPI equations for estimating glomerular filtration rate among patients with varied clinical presentations. *Clin J Am Soc Nephrol*. 2011;6:1963–1972.

dividing the whole-kidney GFR by the number of functional (nonsclerosed) glomeruli in both kidneys allows for an estimation of the mean snGFR.[142] The snGFR remains stable with age because the loss of nephrons parallels the loss in total GFR.[88] Thus the effect of age on snGFR is minimal, consistent with other physiologic characteristics, including height and sex, whereas obesity, family history of ESKD, and nephrosclerosis exceeding that expected for age are associated with an increased snGFR.[142,143]

Although other studies of snGFR in living human subjects are lacking, some studies have indirectly assessed the single-nephron ultrafiltration coefficient (snK_f) in individuals similar to living kidney donors.[113,144,145] The snK_f represents the filtering capacity of the glomerulus, as determined by the surface area and permeability of the glomerular filtration barrier. In these studies, snK_f was estimated from electron microscopy measures of the glomeruli on kidney biopsy.[113] The snGFR could be calculated by multiplying snK_f by the perfusion pressure across the glomerular filtration barrier.[146] Thus it is possible that the relationship of clinical characteristics to snGFR may be similar to that of snK_f. Consistent with the findings of a relatively stable snGFR across the age spectrum described earlier, snK_f did not show significant differences between younger and older living kidney donors.[113,147] Adaptive hyperfiltration occurs in the aging kidney following unilateral nephrectomy[148]; however, either due to an increase in Kf, glomerular capillary filtration pressure, or increased glomerular plasma flow or some combination of the three, the increase in snGFR (and whole-kidney GFR [wkGFR]) is reduced compared with younger persons. The definition of hyperfiltration based on measurements of the whole-kidney GFR is best determined by an age-adapted measured GFR (not eGFR), uncorrected for body surface area. However, such definitions do not distinguish between a high snGFR a or high nephron number.[149]

OTHER FUNCTIONAL CHANGES

RENAL PLASMA FLOW

Renal plasma flow (RPF) represents the volume of plasma delivered to both kidneys per unit of time. The RPF can be calculated from the amount of plasma that is cleared of para-aminohippurate (PAH) per unit of time. PAH is a compound with a nearly 100% extraction through the kidneys. A Swedish study of 122 potential kidney donors, ages 21 to 67, has shown that similar to the GFR, the renal plasma flow (RPF) declines with older age.[143] Older age influences the degree of renal blood flow change as a response to exercise[150,151] and attenuates the renal hemodynamic response to atrial natriuretic peptide[152] and the degree of vasodilation.[153] One study has postulated that higher levels of the endogenous nitric oxide (NO) inhibitor asymmetric dimethylarginine (ADMA) lead to a reduced RPF and hypertension with older age.[154] Animal studies have provided some evidence that ADMA is associated with tubulointerstitial ischemia and fibrosis,[155] glomerular capillary loss, and glomerulosclerosis.[156] In a study of healthy individuals, infusion of dopamine and amino acids was used to test the effects of maximal vasodilation on RPF and NO. Whereas both RPF and NO levels increased in young and middle-aged individuals, they did not change in older persons.[157] This suggests that due to advanced age-related vascular changes in older adults, renal blood vessels are less responsive to maximal vasodilation.

FILTRATION FRACTION

The filtration fraction (FF) represents the proportion of fluid entering the kidneys that reaches into the renal tubules—that is, it is a ratio between the GFR and RPF. Normally, the value for FF is about 20% and remains relatively constant in older age. However, in very old age, atherosclerosis and arteriosclerosis of the arteries and arterioles reduce blood flow to kidneys, thereby increasing FF.[143]

RENAL RESERVE IN AGING

The phenomenon of renal reserve refers to an acute increase in GFR (usually by 20% or more from basal values) when human subjects (or experimental animals) are fed oral protein loads (typically cooked red meat) or infused intravenously with certain amino acids.[157-161] The mechanisms underlying these physiologic phenomena are not fully understood but appear to be a consequence of vasodilatation, an increase in glomerular plasma flow, and an increase in the snGFR[162] (see also Chapter 3). Growth hormone may be involved, but this is controversial.[157-159]

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The efficiency of renal reserve is blunted to some degree by aging per se, most often in the very elderly.[157] Functional reserve is well maintained up to about age 80 years in otherwise healthy men and women.[163] Diminished levels of sex hormones may play a role in the diminished renal reserve seen with aging (see later). The diurnal variation in GFR is also blunted with aging.[164]

Aging in indigenous Kuna Indians (Panama), who subsist on a low-protein, high–dark chocolate diet and remain largely free of hypertension and cardiovascular disease (CVD), is associated with a steady decline in GFR and RPF, similar to that observed in westernized countries.[165] These findings suggest that hypertension is a *consequence* of CKD and/or nephron underendowment, not the cause of CKD with aging.

SEXUAL DIMORPHISM IN AGING

In a series of studies, mainly focused on murine species, Baylis and coworkers[166-170] developed the concept of sexual dimorphism in renal aging (also see review by Gava and associates[171]). Women develop less age-related decline in kidney function, perhaps due to a renoprotective action of estrogens. In contrast, men have more pronounced loss of kidney function with advancing age, perhaps due to the adverse effects of androgens. The mechanisms involved are complex but seem to implicate disturbances of NO synthesis pathways, with a lower rate of NO production potentially influenced by sex steroids.[167,168] A role for the renin-angiotensin-aldosterone system (RAAS) has also been described.[170] Interestingly, in aging experimental animals, there was no observed rise in glomerular capillary pressure or glomerulomegaly in males or females, castrated or not. These findings suggest that the age-dependent sexually dimorphic loss of kidney function does not appear to result in compensatory hemodynamic changes or glomerular hypertrophy, as is observed in humans.[170] Age-dependent sexual dimorphic effects on kidney function might be related to increased body size in males and therefore may be conditional on the balance between energy demand and supply. These findings in murine species have been recapitulated in humans by Melsom and colleagues[172] in a longitudinal study of measured GFR in men and women aged 50 to 62. The measured GFR in healthy women declined more slowly than in men.

Pathophysiologic explanations for aging-associated nephrosclerosis

Podocentric versus ischemic hypotheses (animal and human studies). Evidence indicates that renal blood flow declines with advancing age.[173] Studies in rats have shown that in addition to the reduction in blood flow, renal arterioles in older rats have altered sensitivity to NO[174] and angiotensin II.[175] However, the main limitation of animal studies is the virtual absence of arteriosclerosis, thus limiting studies of age-related ischemic changes. Studies in human kidneys have demonstrated that age-related arteriolosclerosis and intimal and medial hypertrophy in intrarenal arteries resemble those observed in extrarenal arteries.[176,177] These pathologic changes are especially common in interlobular arteries and are frequently observed in kidney biopsies from normal individuals without CVD. Although it is unclear what causes the vascular changes in intrarenal arteries, Martin and Sheaff[125] have postulated that they are directly linked to global glomerulosclerosis, which occurs first in the superficial cortical layers and is followed by local interstitial fibrosis and tubular atrophy (IFTA). Subsequently, as a compensatory mechanism, the deeper glomeruli hypertrophy and, over time, undergo hyperfiltration injury, ultimately developing glomerulosclerosis. More recently, studies in healthy living kidney donors have supported the hypothesis of Martin and Sheaff that age-related glomerulosclerosis is primarily of vascular origin due to observed ischemic changes in the glomeruli[178] and increased prevalence of arteriosclerosis.[92] Moreover, implantation kidney biopsies of older but otherwise healthy kidney donors can also have ischemic-appearing, presumably still functional glomeruli, with a wrinkling of capillary loops, thickened basement membrane, and mild intracapsular fibrosis. These findings may be associated with ischemia and gradually lead to progressive shrinkage of the glomerular tufts and collagen deposition in the Bowman space, ultimately leading to global glomerulosclerosis.

Alternatively, *podocyte depletion* can also give rise to the glomerulosclerosis of aging. The podocytes are highly differentiated, specialized epithelial cells in a glomerulus, with limited capacity for cell division and turnover.[179,180] The two main reasons that lead to podocyte depletion, in absolute or relative terms, are actual loss of podocytes (cell death or detachment from the tuft) and glomerulomegaly—each podocyte subsequently has to cover the increased filtration surface area. The notion that podocyte dysbiosis may be the culprit for glomerulosclerosis is not new. Three decades ago, Kriz and colleagues[181] postulated that podocyte depletion and a subsequent bare glomerular basement membrane are the first steps that lead to focal and segmental glomerulosclerosis. Studies in rats have provided evidence that reduced numbers of podocytes lead to glomerulosclerosis.[182,183] Using puromycin aminonucleoside (PAN) treatment as a model for podocyte oxidative injury, one study has linked glomerulosclerosis and podocyte number depletion.[182] Others have shown increased glomerulosclerosis (usually of the focal and segmental variety) when podocytes fail to follow glomerular growth as a result of molecular growth signaling.[183] Increased glomerulosclerosis may also be due to podocyte hypertrophic stress associated with glomerulomegaly in rats without calorie restriction.[184] Interestingly, calorie restriction abolished glomerulomegaly, podocyte hypertrophy/stress, and podocyte loss with resulting glomerulosclerosis.[184] Using a transgenic approach in rats, the same group has also developed evidence that the reduced number of podocytes alone is sufficient to cause glomerulosclerosis.[185]

Taken together, animal (murine) studies favor the concept that acquired or congenital podocytopenia (absolute or relative) is a critical driving force for the development of glomerulosclerosis, particularly of the focal and segmental variety (see review by Kriz and associates[181]). However, the biology of aging in experimental animals differs from that of humans (see earlier), and these differences must be considered when translating pathophysiologic concepts from animals to humans.

More recently, human kidney biopsies have been used to study podocytes; these have found that the pattern of global glomerulosclerosis with aging is somewhat

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different in studies not limited to living kidney donors.[109] A study of 89 kidney biopsies from living kidney donors, deceased kidney donors, and normal poles of radical nephrectomy specimens performed for tumors found that podocyte depletion is identified as a potential culprit for human age-related glomerulosclerosis.[109] The authors counted podocytes, measured their size and density, and demonstrated a decline in podocyte density with older age due to a reduced count of podocytes per glomerulus and larger glomerular volume. In particular, younger individuals had more than a threefold higher podocyte density compared with individuals older than 70 years (>300 podocytes/10^6 μm^3 vs. <100 podocytes/10^6 μm^3). In the older individuals, the podocyte density was lower and their detachment rate was much higher, and podocytes showed molecular evidence of cell stress. Moreover, in some glomeruli with significant podocyte detachment, the authors observed binucleated podocytes as evidence of failed mitotic attempts and glomerular capillary wrinkling, tuft collapse, and pericapsular fibrosis. These findings led the authors to propose a hypothesis (the podocentric hypothesis) for the glomerular natural history with aging, extending from glomerular hypertrophy to tuft collapse to glomerulosclerosis.[109] However, studies limited to living donors did not show glomerular hypertrophy with aging,[86] suggesting that age-related comorbidities rather than aging alone may explain these findings. In addition, it is noteworthy that albuminuria is not a feature of normal healthy aging.[92] Thus the podocentric hypothesis may not be relevant to human renal aging.

CONTRIBUTION OF COMORBIDITIES TO RENAL AGING

Aging is commonly accompanied by comorbidities that can have an independent effect on kidney structure and function. These include obesity, type 2 diabetes, and nephron endowment at birth. For example, obesity and/or type 2 diabetes can lead to glomerulomegaly, glomerular hyperfiltration, and albuminuria (see Chapter 50). In part, these comorbidities contribute to the rising prevalence of albuminuria with aging seen in cross-sectional epidemiologic studies.[69] Hypertension, commonly seen in older adults, can also modify the underlying pattern of renal senescence, as discussed earlier.

It is likely that nephron endowment at birth modifies the pace of normal physiologic renal aging. Low nephron endowment at birth, due to fetal dysmaturity and/or undernutrition, will be accompanied by early single-nephron hyperfiltration and glomerular enlargement, which can lead to maladaptive glomerular injury, podocytopenia, glomerulosclerosis, and perhaps an acceleration of the normal rate of nephron loss.[186] The combination of low nephron endowment at birth and physiologic renal senescence could, and likely does, lead to features of CKD (hypertension, reduced GFR, and/or albuminuria) observed in later life. Reduced glomerular density and glomerular hypertrophy are signs of nephron underendowment and have been shown to accelerated progression of many kidney diseases, including some that typically affect older adults.[187,188] Low birth weight, fetal dysmaturity, and impaired nephrogenesis can contribute to the burden of kidney disease globally, especially in older adults and in persons with a single kidney (surgical or congenital) or primary/secondary causes of CKD (see Chapter 20).[188]

FLUID, ELECTROLYTE, AND ACID-BASE HOMEOSTASIS IN AGING

SODIUM HOMEOSTASIS

Studies in healthy human subjects clearly demonstrate that kidney aging reduces the capacity of the tubules to conserve filtered sodium, probably by effects in the distal nephron, previously discussed in Chapter 13.

Sodium homeostasis can also be influenced by age-related changes in levels and responses to renin and aldosterone, hormones that regulate sodium conservation. Although plasma renin activity and aldosterone levels are generally lower in older adults, they are not associated with changes in fluid or electrolyte metabolism.[189] These changes increase older adults' susceptibility to sodium retention in disease states or with the use of medications that alter renin release.

It has also been hypothesized that the higher prevalence of hypertension in older persons may be due in part to the attenuated natriuretic response with aging. There is also a reduction in the natriuretic response to saline loading in older versus younger kidney donors after donation.[190]

Atrial natriuretic peptide (ANP) is a hormone secreted by atrial myocytes, and one of its functions is to control sodium excretion at the tubular level. With older age, the tubular response to ANP is reduced. ANP inhibits sodium reabsorption from the luminal side and induces hyperfiltration and suppression of renin release, which together lead to natriuresis, diuresis, and lowering of blood pressure. Studies have shown that the plasma ANP concentrations are several times higher in older individuals, and this may be a compensatory effect of the reduced response at the receptor level. This hypothesis is in agreement with two studies in which sodium excretion plateaued after an ANP infusion in older subjects,[191,192] whereas in younger subjects sodium excretion continued to increase with increasing ANP dose.[191] Of note, it appears that ANP production does not change with older age; therefore the observed increase of ANP in older adults results from reduced metabolic clearance.[193,194]

The role of altered renal sodium handling in the pathogenesis of age-related elevation of systemic arterial pressure is a topic of great contemporary interest (see also Chapter 46; reviewed by Frame and Wainford[195]). Vascular factors, such as reduced compliance of major arterial vessels, and neurohumoral alterations, such as increased sympathetic nervous system activity, play important roles in the elevation of systolic blood pressure with aging. The sensitivity of blood pressure changes consequent to salt administration increases with older age.[196]

The expression of the epithelial sodium channel (ENaC) can be reduced in aging rodents.[197] These studies collectively indicate that dysregulation of sodium handling by the aging nephron can be involved in age-related blood pressure elevation. Integration of this knowledge into the complex array of biologic factors in systems affecting blood pressure (e.g., aortic compliance, sympathetic nervous system activity, RAAS) will be challenging.[195] Moreover, the role of reduced nephron mass that accompanies physiologic aging cannot be ignored.

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POTASSIUM HOMEOSTASIS

Potassium homeostasis is regulated by several different mechanisms; specifically, in the kidney, the potassium excretion rate is altered depending on the intake (discussed further in Chapter 16).[198]

Total body potassium (TBK) declines with older age with a reduction in intracellular stores that accompany the decline in muscle mass (sarcopenia) seen with normal aging. The observed decline in TBK may also be explained by the lower renin and aldosterone levels,[199] as well as the blunted response to aldosterone in older adults (see later).[200] Alternatively, older adults may express impaired ability to excrete acute potassium loads. Insulin-mediated potassium homeostasis in human subjects is not influenced by age.[201]

Older age is often accompanied by comorbid conditions for which medications that interfere with potassium excretion might be prescribed. Examples include potassium-sparing diuretics, angiotensin-converting enzyme (ACE) inhibitors, angiotensin receptor blockers, heparin, calcineurin inhibitors, sodium channel blockers, and nonsteroidal antiinflammatory drugs (NSAIDs). Care is needed when starting these medications for older patients, given the risk of hyperkalemia.

Clinical Relevance

Older adults are at increased risk of hyperkalemia, even given their reduced total body potassium stores and altered tubular potassium transport.

MAGNESIUM HOMEOSTASIS

It is estimated that about 99% of magnesium is stored in cells, predominantly in bones and muscles, and only about 1% is in extracellular fluid including plasma. Normally, the kidneys reabsorb 96% of filtered magnesium[202] and regulate magnesium homeostasis (see Chapter 17).

Because plasma and intracellular magnesium homeostasis are finely regulated,[203] total plasma magnesium is maintained remarkably stable with increasing age in healthy individuals.[204,205] However, subclinical deficits of magnesium not detected by serum magnesium concentrations can occur in older individuals.

Changes in magnesium homeostasis seen in the elderly are likely due to diseases that frequently occur in older age, use of therapeutic agents that alter its metabolism, and an age-related decline in kidney function.[204] On the other hand, hypermagnesemia occurs rarely, even in the disease setting, including in those with severe acute kidney injury (AKI), CKD, or ESKD.[206] Interestingly, low dietary magnesium intake in older adults may also contribute to a faster rate of decline in GFR, potentially exacerbating a proinflammatory milieu, endothelial dysfunction, and promotion of vascular calcification.[207]

CALCIUM HOMEOSTASIS

In normal adults, approximately 10 g of calcium is filtered through the glomeruli each day, and 98% to 99% is reabsorbed. Net calcium excretion during a 24-hour period ranges from 100 to 200 mg. Most filtered calcium reabsorption (≈60%–70%) occurs in the proximal tubules, around 20% in the loop of Henle (thick ascending limb), around 5% to 10% in the distal convoluted tubule, and only around 5% in the collecting ducts (see also Chapter 17).

Calcium homeostasis with aging has been extensively studied in man and experimental animals. Some animal and human studies have shown that there are age-related changes in renal calcium homeostasis, such as decreased tubular reabsorption and/or the renal response to parathyroid hormone (PTH),[208-211] despite relative preservation of renal calcium excretion with aging.[207] Renal and intestinal calbindins are proteins with a critical role in active transcellular calcium transport, and their expression decreases with age.[212-214] There is also an age-related decrease in the capacity of 1,25-dihydroxyvitamin D_3 (1,25-[OH]$_2$$D_3$) to stimulate calcium reabsorption, indirectly increasing PTH concentrations in both rats and humans.[215,216] Deficits in renal and duodenal calcium intracellular transporters can lead to impaired renal and duodenal calcium reabsorption, consistent with the increased calciuria observed in older mice.[217] Alpha-Klotho (α-Klotho) is a gene associated with accelerated aging when mutated.[218] It encodes a protein that is predominantly expressed in tissues involved with calcium homeostasis. A study in mice has demonstrated that alpha-klotho binds to a subunit of sodium-potassium adenosine triphosphatase (Na^+-K^+-ATPase),[219] suggesting that the age-related decline in Klotho protein could contribute to reduced sodium-potassium ATPase sensing of low calcium levels. Serum-soluble α-Klotho (sKlotho) levels decline with advancing age in humans, independent of the GFR.[55]

There is also an age-related decrease in intestinal calcium reabsorption, which is linked with reduced 1α-hydroxylase activity, lower serum 1,25-(OH)$_2$$D_3$ concentrations, and higher serum PTH concentrations. Moreover, older men also had a higher threshold for PTH release; however, this was not associated with a decrease in blood ionized calcium or 1,25-(OH)$_2$$D_3$.[220] This finding suggests that the relationship between PTH and calcium in older age is changed so that irrespective of calcium concentrations, serum PTH concentrations are higher. An interesting study has compared healthy older individuals (≥75 years of age) to patients with CKD and a similar GFR and found that patients with CKD have higher fractional excretion of calcium, suggesting more pronounced calcium wasting in these patients.[221] Concentrations of vitamin D–dependent calcium-binding proteins also decrease with older age, which is associated with the change in intestinal calcium absorption.[212,222] Another study found that the renal response to PTH infusion and renal vitamin D levels were equivalent in both young and older adults, with the only difference being that the production of vitamin D is somewhat delayed in older adults.[223]

PHOSPHORUS HOMEOSTASIS

Decline in the GFR with older age leads to reduced levels of non–protein-bound (free) serum calcium but increased levels of serum phosphate.[224] In response to these changes, PTH synthesis increases, which in turn decreases the number of sodium phosphate cotransporters in the proximal tubules, thus leading to increased phosphaturia. Elevated serum

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phosphate concentrations also stimulate the production of FGF23, which regulates serum phosphate concentrations in the setting of impaired kidney function and suppresses production of 1,25 $(OH)_2D_3$. FGF23 concentrations have been found to increase with diminished GFR but not with age per se.[55]

In addition to the hormonal regulation of serum phosphate concentrations in human aging, animal studies have demonstrated a PTH-independent reduction in the intrinsic tubular capacity to reabsorb phosphate.[225] Others have demonstrated that kidneys in aged animals show impaired phosphate transport in renal tubules in response to a low-phosphate diet.[226,227] It has been proposed that the reason for this impaired phosphate transport is decreased fluidity of a luminal brush border membrane due to the age-related increase in cholesterol and sphingomyelin,[227] which then diminishes sodium phosphate cotransporter activity directly.[228] Consistent with in vivo studies, an in vitro study of cultured renal tubular cells harvested from young and adult rats showed reduced phosphate uptake and adaptation to a phosphate-free medium only in the adult kidney cells.[229] Another study revealed that the reason for the age-related decline in tubular phosphate reabsorption and tubular adaptation to a low dietary phosphate is lower expression of renal sodium phosphate cotransporters on the apical brush border membrane.[230] As mentioned earlier, age-related reduction of α-klotho (a receptor for FGF23) may also be involved in the changes in phosphate hemostasis seen with aging.

VITAMIN D HOMEOSTASIS

Older persons commonly exhibit vitamin D insufficiency or deficiency, often in association with a subtle state of chronic inflammation. Serum concentrations of 25-hydroxyvitamin D (25- [OH] D) are inversely associated with biomarkers of inflammation in older adults, such as C-reactive protein and interleukin 6 (IL-6).[231] It is not known whether these changes are due to primary changes in vitamin D metabolism or a secondary effect of inflammation. Vitamin D absorption from dietary sources may not be impaired with aging, but reduced dietary intake can contribute to low serum 25-(OH) D levels in aging individuals.[232] Reduced production of 7-dehydrocholesterol in the aging epidermis with ultraviolet light exposure is likely involved in the decreased production of pre–vitamin D_3 (cholecalciferol) in older adults.[233] Serum 1,25-$(OH)_2D$ (calcitriol) concentrations decline in aging, at least in part due to the decline in the GFR and impaired hydroxylation of 25-(OH)D mentioned earlier.[232] Changes in vitamin D sufficiency or insufficiency in older persons can modulate left ventricular geometry and modulate the risk of left ventricular hypertrophy in hypertensive older subjects.[234]

It is important to keep in mind the age-related changes of 1,25-$(OH)_2D$ metabolism on intestinal phosphate absorption (described earlier) because several studies have shown that dietary vitamin D supplementation improves renal and intestinal phosphate absorption in vitamin D–deficient animals.[235-237]

Thus aging has diverse effects on calcium, phosphorus, and vitamin D metabolism. These include a decrease in intestinal absorption of calcium, impaired vitamin D production in the skin, dietary vitamin D deficiency, impaired production of 1,25-$(OH)_2D_3$, increased PTH secretion, lower serum Klotho concentrations, higher serum FGF23 concentrations, and alterations in renal tubular handling of calcium and phosphorus. Interestingly, low serum Klotho concentrations also appear to increase the risk of hyperuricemia.[238]

ACID-BASE HOMEOSTASIS

The kidneys play a major role in the regulation of acid-base homeostasis. Normally, during a 24-hour period, kidneys reabsorb around 4500 mmol of filtered bicarbonate. Kidneys have a large capacity to excrete the endogenously generated proton (H^+) by generating ammonia (NH_3) and its excretion (NH_4^+; see Chapter 15). Older adults are more prone to develop acid-base dysregulation for several reasons. First, age-related structural and functional changes in the kidney reduce the adaptive responsiveness to dietary and/or environmental changes. Second, the age-related decline in GFR reduces the capacity of the kidney to buffer metabolic changes and excrete the excess H^+ load.[239] Along with age-related decline in kidney function, there is also a reduced capacity for bicarbonate conservation and generation, which, combined with the stable endogenous H^+ production, may lead to metabolic acidosis. This process is enhanced in patients with concomitant CKD.[240] A community-based observational study of healthy persons aged 6 to 75 years has found that the capacity for H^+ excretion is similar from childhood to young adulthood but significantly reduced in older age.[241]

Ammonium excretion also decreases with older age.[242] Taken together, it has been suggested that the age-related decrease in H^+ excretion is due to reduced renal tubular mass rather than tubular dysfunction.[242] Overall acid production comes from two sources, namely the oxidation of sulfur atoms on amino acids and metabolically produced organic acid ions. In a study of these parameters in aging, Huo and colleagues[243] showed that acid production rates increase with age.

Overall, these studies suggest that older age appears not to alter the fundamental physiologic regulation of acid-base homeostasis but does lead to reduced and diminished responses to acid-base challenge. Consistent with this, a study of older patients with CKD has found that oral sodium bicarbonate supplementation corrects metabolic acidosis, increases serum albumin concentrations, and decreases whole-body protein degradation.[244] Potassium bicarbonate supplementation in postmenopausal women neutralized the endogenous acid, improved calcium and phosphate balance, and reduced bone reabsorption.[245] This was confirmed in a double-blind study of older men and women—in which particularly bicarbonate and not potassium—had a protective effect on bone reabsorption and calcium excretion.[246] Taken together, the results of several studies are consistent with the hypothesis that a higher alkaline residue (vegetarian) diet prevents bone loss in older individuals.

WATER HOMEOSTASIS

Water homeostasis is regulated by a high-gain feedback mechanism that involves the hypothalamus, neurohypophysis, and kidneys.[247,248] In the kidneys, water and sodium from the glomerular filtrate

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are reabsorbed in tubules through water channel aquaporins (AQPs) and sodium cotransporters.[249-251] The tubular reabsorption of water depends primarily on the driving force (high interstitial osmolality in the deeper medullary zones) and osmotic equilibrium of water across the tubular epithelia (high osmotic water permeability of the membrane). The large majority of glomerular filtrate is reabsorbed in the proximal tubules and descending thin limbs of the loop of Henle.[250,252] The next tubular segments (thin and thick ascending limbs and distal convoluted tubules) are relatively water impermeable.[253,254] Finally, the main parts of the nephron where vasopressin-based regulation of body water homeostasis occurs are the connecting tubule and collecting duct[255-257] (see also Chapter 14).

Maximum urinary concentration and diluting capacity are decreased with aging,[258] leading to a higher risk for hypernatremia and hyponatremia. Findley has proposed an age-related dysfunction of the hypothalamic-renal axis based on clinical observations of increased vasopressin secretion in older age.[259] With older age, maximal urine-concentrating ability is particularly reduced.[258] Compared with younger individuals, older adults have about a 50% reduced capacity to conserve water and solutes. One of the major changes in body composition with aging is an increase in total fraction of body fat by 5% to 10% and an equivalent decrease in total body water, so an older man on average has 7 to 8 fewer L of total body water than a young man with the same body weight.[260] The obvious consequence is that in the event of acute loss, or overload of body water, a more severe change in serum osmolality will occur in older individuals. A study comparing plasma osmolality in older and younger individuals before and after a similar extent of water deprivation has confirmed this presumption.[261] Alterations in water and sodium balance frequently lead to hyponatremia or hypernatremia associated with hypovolemia or hypervolemia in older adults.[262,263] In addition, dysfunction in water homeostasis may also occur due to abnormal expression and trafficking of AQPs and solute transporters. For example, an animal study has shown that the AQP2 transporter involved in urine-concentrating ability is downregulated in the medulla of older rats.[264] This molecular mechanism is consistent with a reduced urine-concentrating ability in older adults. Water homeostasis is also influenced by the previously described microstructural changes that occur with the aging kidney. In the event of stress, acute disease, volume load, or dehydration, the combined effects of age-related loss of renal mass (i.e., functional reserve) and the change in body composition may cause a significant disruption in water and solute homeostasis.[265,266]

One of the consequences of the age-related GFR decline is an increase in filtrate reabsorption in the proximal tubules and decreased fluid delivery to the distal diluting tubules, resulting in a reduced diluting capacity of the kidney.[267] This is evidenced by a reduced ability to excrete a free water load[261] or reduced maximal free water clearance in older adults.[267] In addition to the reduced diluting capacity, older kidneys also lose the capacity to conserve water during states of dehydration.[268] Taken together, therefore, these age-related changes may have significant clinical implications, such as worsening of dehydration in older adults in the setting of vomiting, diarrhea, or reduced water and food intake.

Vasopressin secretion, the renal response to vasopressin, and thirst control are also affected by aging. Vasopressin secretion in the hypothalamus is under the delicate control of osmoreceptors in and around the organum vasculosum of the lamina terminalis and the anterior wall of the third ventricle in the brain. Most, but not all, studies have found that basal vasopressin concentrations in the healthy older are usually higher than in younger controls.[260,269,270] Nevertheless, most studies about water homeostasis in aging have shown that compared with younger individuals, older adults have a greater increase of vasopressin secretion per unit change in plasma osmolality, and this is consistent with the higher osmoreceptor sensitivity in older adults.[271]

Vasopressin regulates renal water excretion by controlling the abundance of the water channel AQP2 and its insertion into the apical membrane of epithelial cells in the distal nephron and collecting tubules.[272] In the membrane, AQP2 proteins form channels that allow reabsorption of water molecules from the lumen of the collecting ducts into the medullary interstitium driven by the medullary osmotic gradient. Given that vasopressin levels are mostly found to be higher in older adults, a potential secretory defect in the pituitary gland is improbable and cannot explain the age-related reduced renal response to vasopressin. Animal studies have offered potential explanations for the reduced renal response to vasopressin, which include reduced expression of vasopressin receptors in the collecting ducts and impaired second messenger response to vasopressin receptor signaling.[273,274]

Normally, stimulation of thirst osmoreceptors in the hypothalamus signals the higher cerebral cortex to develop a conscious perception of thirst- and water-seeking behavior. Aging also affects thirst control.[275-277] It has been proposed that older adults may have a higher osmolal set point for thirst—that is, for a given plasma osmolality, older adults have a reduced degree of perceived thirst, thus leading to less water intake.[278]

Thus aging undoubtedly influences water homeostasis in many ways. Water conservation (urinary concentrating ability) is primarily affected, although a mild form of impaired diluting capacity may also be present. This is important to keep in mind during the diagnostic process and while taking care of older adults and planning for different clinical, surgical, or pharmacologic interventions. Alterations in solute intake (e.g., protein or sodium) in older adults can also have profound effects on water homeostasis.

RENAL ENDOCRINE CHANGES WITH AGING

Renin-Angiotensin-Aldosterone System

Aging is associated with a decline in plasma renin activity (PRA), serum aldosterone concentrations, and urinary aldosterone excretion rates, accompanied by reduced responses to actions that stimulate the RAAS, such as upright posture or salt depletion.[62,63] This state is known as *aging-associated hyporeninemic hypoaldosteronism*. Both renal renin formation and release are reduced in older individuals.[279] In addition, activity of the RAAS has been linked to aging phenomena in a pathogenic fashion.[63] Defects in renal NO generation can be linked to reduced activity of the RAAS

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in aging.[11] The activity of the RAAS can also be linked to canonical Wnt signaling,[11] and both can be dysregulated in the aging process.[9] Although a decline in activity of the RAAS regularly occurs with aging (independent of sex), neither a change in renal perfusion pressure nor delivery of solute to the macula densa appears to be involved.[11] Increased serum ANP concentrations, reduced sympathetic nervous system activity, or both can be seen with aging and might be involved in the perturbations of the RAAS seen with aging.[11] Reduced renal renin generation with aging appears to be a posttranslational phenomenon due to impaired release of stored renin.[11] The sensitivity of the glomerular afferent and efferent vessels to angiotensin II appears to increase with aging.[11] Distinguishing the effects of aging per se and superimposed disease states (e.g., low nephron endowment) on the RAAS can be difficult.

Aldosterone production and serum and urinary aldosterone levels are lower in older adults compared with younger persons,[280] and there may also be reduced aldosterone responsivity to elevations in serum potassium levels.[200] In addition, there are age-related changes in adrenal aldosterone synthetase–producing cells, causing islands or clusters of aldosterone-producing cells in the adrenal cortex of aging humans. These may be precursors of aldosterone-secreting adenomas in some cases.[280]

ERYTHROPOIETIN

Serum levels of erythropoietin (EPO) tend to increase with age, even in nonanemic subjects.[200,281] This might be in compensation for increased (subclinical) blood loss, accelerated red blood cell turnover (decreased erythrocyte half-life), or increased resistance of red cell precursors to the effect of EPO.[281] Testosterone deficiency might be involved in the latter phenomenon.[282] Mild anemia (hemoglobin <12 g/dL) is fairly common in older persons but varies by geography.[283] In about one third of persons with anemia, a nutritional deficiency (e.g., iron, folate, vitamin B_{12}) was responsible; in one third, a concomitant chronic disease was the culprit (e.g., CKD, cancer, and infection); and in one third, no precise cause could be determined.[283] Independent of anemia, high spontaneous EPO levels in older adults are associated with an increased risk of congestive heart failure but not myocardial infarction or progressive CKD.[284] The mechanisms responsible for this association are not well understood but could be due to tissue hypoxia in low-perfusion states associated with heart failure.

IMPLICATIONS OF AGE-RELATED CHANGES IN NORMAL PHYSIOLOGY ON CHRONIC KIDNEY DISEASE DIAGNOSIS AND PROGNOSIS

DIAGNOSIS

The definitions and classifications of CKD have been codified by the Kidney Disease Outcomes Quality Initiative (KDOQI) and Kidney Disease: Improving Global Outcomes (KDIGO) clinical practice guideline initiatives,[285,286] published in 2002 and 2012 and revised in 2023, as well as others from more regional formulations, such as NICE in the United Kingdom and CARI in Australia.[287,288] These recommendations rely principally on the GFR (estimated or measured, in mL/min/1.73 m^2) and albuminuria (usually the urinary albumin-to-creatinine ratio [uACR], in mg/g or mg/mmol] in spot urine samples. However, other signs of kidney injury, such as abnormal urinalyses (e.g., hematuria), imaging, or kidney biopsy findings, could lead to a diagnosis of CKD, even with a normal GFR (even adjusted for age) or uACR. Abnormalities in these biomarkers must persist for at least 3 months to qualify as a valid indicator of CKD, a requirement that is frequently not met in many epidemiologic studies. When these classification schemes were developed, it was arbitrarily decided that any adult 20 years of age or older, with a persisting GFR (measured or estimated) less than 60 mL/min/1.73 m^2, could be classified as having CKD (category 3a, 3b, 4, or 5), irrespective of any other signs of kidney injury including abnormal urinalysis, albuminuria/proteinuria, imaging, or renal pathology. Thus a persisting GFR of 45 to 59 mL/min/1.73^2 could lead to a diagnosis of CKD in any adult, irrespective of age, even in the absence of albuminuria <30 mg/g (category G3a/A1). This aspect of the KDIGO guidelines created a conundrum because values of the GFR in this range can also be observed in healthy older adults. The lower limit of normal (fifth percentile) for the measured GFR (mGFR) in a healthy adult 65 to 70 years of age is 56 to 60 mL/min/1.73 m^2 and falls progressively: for 70 to 75 years to 52 to 56 mL/min/1.73 m^2, for 75 to 80 years to 47 to 52 mL/min/1.73 m^2, for 80 to 85 years to 43 to 47 mL/min/1.73 m^2 and for 85 to 90 years to 34 to 42 mL/min/1.73 m^2.[133,289] Thus a sizable overlap is present between the single (absolute) threshold for defining and classifying CKD (independent of age) and normal values for mGFR in older adults (older than age 65 years). The lower value for the GFR in older adults is a manifestation of a normal physiologic process of nephron loss unaccompanied by compensatory hyperfiltration in the residual nephrons (see earlier), so it is difficult to conflate these findings as a disease. Not surprisingly, a substantial and continuing debate has arisen concerning the utility of an absolute threshold for defining CKD, based on the measured or estimated GFR alone, without any modification for age.[290-294] Because abnormal albuminuria is not consistent with normal, healthy human aging, this controversy focuses largely on overdiagnosis in older adults of the G3a/A1 category of CKD.[295] The GFR may fall below 45 mL/min/1.73 m^2 in subjects older than 85 years, but it can be difficult to determine if such persons are entirely healthy because they frequently have concomitant disorders, such as cancer, heart failure, protein energy malnutrition, and sarcopenia. These conditions can complicate evaluation of the GFR (mGFR or eGFR). In addition, significant comorbidity and biochemical abnormalities are commonly observed in individuals older than 65 years of age, compounding the difficulty of ascertaining kidney health based on the GFR alone.[296,297]

Typically, GFR values used in categorizing CKD are estimates rather than measured values. This use of estimating formulas for the GFR (eGFR) results in another conundrum in the diagnosis of CKD in aging—namely, the imprecision and variability in eGFR formulas for evaluating the true GFR in older adults[296] (discussed further in Chapters 23 and 60). In a population-wide study of community-living adults, Ebert and coworkers[296] found the prevalence of CKD categories 3,

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4, and 5 (eGFR <60 mL/min/1.73 m^2) in subjects 70 to 79 years of age to range between 16% and 52%, depending on the formula used to estimate the GFR. The corresponding values for subjects 80 to 89 years of age were 42% to 84%. The lowest values for CKD prevalence were found using the CKD-EPI-creatinine equation (2009 version); the highest values were found using the Berlin Initiative Study (BIS)-1 creatinine equation.[296] In persons older than 70 years of age, the BIS-1 creatinine equation is a more precise estimate of mGFR compared with the CKD-EPI equation, at least in a European, largely white population.[298] Estimating formulas for GFR when applied at the population level generally give high values for CKD prevalence (≈11%–14%),[299] especially when the formulas were derived from a largely CKD cohort. Much lower values for CKD are observed when eGFR formulas are derived from a normal healthy cohort,[300] especially when an age-adapted threshold of GFR is used for defining CKD.

The use of cystatin C–based GFR estimating equations (alone or in combination with creatinine-based equations) has been advanced as a tool to aid in the diagnosis of CKD. In 2012, KDIGO suggested measuring eGFR-creatinine-cystatin C and eGFR-cystatin C in adults (including older adults) when the eGFR-creatinine is 45 to 59 mL/min/1.73 m^2 (and no other markers of CKD are present). If eGFR-cystatin C or eGFR-creatinine-cystatin C are less than 60 mL/min/1.73 m^2, the diagnosis of CKD is confirmed.[286]

It must also be recalled that cystatin C is a mid-molecular-weight (13.3 kD) serine proteinase inhibitor involved in the inflammatory response.[301] It is normally filtered and then completely destroyed by tubular reabsorption and degradation. Little cystatin C is present in normal urine. Thus production rates of cystatin C are impossible to assess (without a simultaneous mGFR), and they can vary in many states (e.g., obesity, diabetes, thyroid disease, inflammation) commonly found in older adults. Like creatinine, there are many non-GFR determinants of its serum concentration, the variable used in calculation of the eGFR.[302,303] In older adults, loss of muscle mass (sarcopenia) may lower serum creatinine values relative to the measured GFR and give rise to an overestimate of mGFR by eGFR creatinine equations. However, serum cystatin C levels tend not to be affected greatly by muscle mass or sex and not at all by ancestry. In contrast, in older adults with chronic inflammation (of any cause), obesity, diabetes, and the metabolic syndrome may alter cystatin C production, leading to underestimation of the mGFR. The combination of eGFR-creatinine-cystatin C may give a slightly more accurate assessment of the true GFR in population-based studies,[304] but there will still be much individual variation.[269,270,305] Tracking eGFR-cystatin C with metabolic factors that can contribute to CVD complicates its suitability as a biomarker for CVD risk related to the GFR in older adults.[273] In addition, endothelial injury consequent to inflammation or capillary hypertension might further impair the transglomerular permeability coefficient for cystatin C (the shrunken pore hypothesis),[274] giving rise to reduced cystatin C eGFR values relative to the inulin or iohexol mGFR.[274] Furthermore, lower serum albumin concentrations (perhaps reflecting subtle degrees of chronic inflammation) are associated with lower eGFR-cystatin C in older frail subjects.[306] The implications of the "shrunken pore" pathophysiology upon the relationships between eGFR-creatinine and eGFR-cystatin C values (especially when they are discordant) need to be more fully explored.[307-309]

Direct comparisons of a gold standard method of measuring the GFR and estimating formulas for the GFR in aging subjects are relatively uncommon, but both the BIS[298] and Reykjavik older cohort studies[269] have provided much valuable information. In the latter study, the Lund-Malmo and full age spectrum (FAS) creatinine equations have somewhat higher accuracy than the CKD-EPI-creatinine formula. The CKD-EPI (2009), Lund-Malmo, and FAS creatinine-based equations were roughly equivalent for the detection of an mGFR less than 60 mL/min/1.73 m^2. All formulas incorporating a cystatin C measurement exhibited consistent improvements in accuracy compared with the corresponding creatinine-based equations. The relevance of these findings to the diagnosis of CKD in individual older subjects needs further study, but they clearly demonstrate some of the pitfalls in using creatinine-based eGFR formulas for the older adult population. Ma and colleagues[310] conducted a systematic review of comparisons of eGFR estimating equations and found that BIS and CKD-EPI-creatinine + cystatin C equations had the best performance characteristics in older adults. When eGFR-creatinine and eGFR-cystatin C are discordant (more than 5 mL/min/1.73 m^2 difference), a fairly common finding in older adults, the lower of the two values or the average of eGFR + eGFR-cystatin C are the more accurate and less biased value relative to mGFR,[311,312] but uncertainty remains over which estimating equation has the best utility for both assessing mGFR and prognosis in elder adults.[313] Investigators have examined the trajectory of repeated measures or estimates of GFR in apparently healthy community-living adults. Schaeffner and colleagues[314] carried out a longitudinal study over 6.1 years in older adults enrolled in the BIS. The eGFR decline depended on both sex and age and tended to decelerate with advancing age. eGFR-cystatin C showed a nonlinear declining slope with age, where eGFR-creat slope was relatively linear. Stability of eGFR-creatinine, even in the range of CKD3b, in the very elderly usually translates to nonprogression, except with concomitant presence of high albuminuria.[315]

Thus the effects of physiologic kidney aging on the GFR confound its use as the defining characteristic of CKD in older adults because CKD is a diagnosis made most frequently in older adults with category 3A CKD (GFR, 45–59 mL/min/1.73 m^2), usually with normal albuminuria. This conundrum raises an issue of overdiagnosis and mislabeling of CKD in the elderly, at both the individual- and population-wide level, that distorts evaluation of the true societal burden of CKD. This is especially true in populations having a high frequency of elderly subjects (such as Japan, Italy, France, and to some extent, China and the United States). When added to the false-positive identification of CKD in single epidemiologic studies (see earlier), this leads to the possibility that CKD is not as common as it is alleged to be and suggests that calls for an age-adapted GFR definition of CKD should be taken seriously. The vagaries of estimating equations for the GFR in older adults cannot be ignored.[316] Use of an age-adapted definition of CKD sharply decreases the overall prevalence of CKD in the population and

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substantially decreases the overdiagnosis of CKD seen with the KDIGO fixed and absolute GFR definition of CKD.[317-320]

PROGNOSIS

In the 2012 KDIGO classification of CKD, the prognosis for adverse events (all-cause mortality, ESKD, doubling of serum creatinine level, CVD) is the third dimension of the scheme. This is usually displayed as a multicolored heat map of the rising risk of events—green, no added risk; and red, high risk—often determined from large-scale epidemiologic studies.[285,286] The risk values are usually stated as relative risk or odds ratio compared with a control group, often those with eGFR greater than 60 mL/min/1.73 m^2 and no signs of kidney disease (e.g., normal uACR). These prognosis- based heat maps are commonly assembled from data acquired from single time point epidemiologic studies, so the persistence of the CKD-defining abnormality (eGFR, uACR, or both) is not confirmed. When a more rigorous assessment of chronicity is pursued, it has been noted that such single time point studies commonly overestimate the prevalence of CKD by as much as 30% due to false-positives.[321]

The use of eGFR to evaluate prognosis is also confounded by the fact that an age variable is included in all adult estimating formulas to adjust for the effects of age per se on the synthesis and production of the requisite biomarker (serum creatinine or cystatin C). The age coefficients in these formulas are optimized for estimating the mGFR, not prognostication for all-cause mortality or ESKD risk. However, when the values obtained by such estimates are applied to prognostication, the age variable becomes important because age, independent of the GFR, has a marked influence on some of the risks evaluated, such as CVD and cancer mortality. If an estimating equation using one or more biomarkers that accurately assess GFR, without a requirement for the age variable, were developed, the accuracy of the risk prediction might be improved and made comparable with that found with the true GFR. It should be stressed that although the number and size of studies using mGFR in prognostic evaluation are relatively small in comparison with the large-scale, eGFR-based epidemiologic studies, the patterns linking GFR to outcome are similar. These studies describe a pattern of a threshold (nonlinear) relationship, in which the threshold varies according to age.[322] With albuminuria, the risk varies in a log-linear fashion with the uACR, independent of age, and without a threshold (above normal values). When one compares the relative risk of all-cause mortality using a reference value for the GFR that approximates the normal young adult mean value (≈107 mL/min/1.73 m^2), then the minimum relative risk for an all-cause mortality event is above an eGFR of 75 mL/min/1.73 m^2 for subjects 18 to 54 years of age, but above 45 mL/min/1.73 m^2 for those older than 75 years.[323] The relative risk of all-cause mortality relative to a declining eGFR is blunted by advancing age, but the absolute rates of all-cause mortality remain higher with aging. This analysis strongly suggests that with a prognosis-dominated matrix for creating a GFR threshold for CKD, an age-stratified approach is desirable. It is noteworthy that the remaining life expectancy at any age older than about 35 years is not materially affected until the eGFR falls below about 45 mL/min/1.73 m^2.[324-326] Adding an age stratification to the existing scheme for identifying true CKD and its associated risks have proven to be a challenging task. Simple adjustments, such as altering the threshold of eGFR to <45 mL/min/1.73 m^2 for subjects without abnormal albuminuria who are older than 65 years, may be useful, but result in unintended consequences, like the so-called birthday paradox (when a 64-year-old reaches a 65th birthday, CKD is "cured").

Other methods of age stratification for CKD diagnosis that may have greater practicality in large health care systems for improved triage have been advanced.[327] Creation of multiple age-stratified heat maps using the eGFR and estimates of prognosis for all-cause mortality may also be helpful.[328] The impact of the specific formula for identifying the eGFR should also be kept in focus. For example, the eGFR-creatinine formulas do not add much to the prediction of CVD (and thereby all-cause mortality) risk in older adults, whereas eGFR-cystatin C formulas do, despite not being superior to eGFR-creatinine for estimating the mGFR.[273]

CARDIOVASCULAR DISEASE AND DECLINE IN THE GLOMERULAR FILTRATION RATE WITH AGING

There is little doubt that a progressive decline in the GFR is associated with an increased risk of fatal or nonfatal CVD, including atherosclerotic CVD and congestive heart failure (CHF), sudden cardiac death, nonvalvular atrial fibrillation, stroke, and peripheral vascular disease. Cross-sectional, population-based epidemiologic studies cannot easily determine whether these associations are causal or not. In fact, the existence of CVD may itself result in a decline in the GFR (e.g., severe CHF, atherosclerotic renovascular diseases, ischemic nephropathy), thus confounding the directional nature of the observed associations. Aging itself can be a mechanism for some of the associations between a decline in the GFR and CVD, but this can be adjusted for by examining the CVD prevalence in older subjects with a better-preserved GFR. However, the fact that the GFR declines with aging requires that the comparator group used for such adjustments has GFR values (on average) similar to those predicted by normal physiologic aging. Teasing out age-related GFR decline from pathologic GFR decline in an aging individual can be difficult; the demonstration of other biomarkers of renal injury, such as the presence of albuminuria or abnormal imaging, can be helpful.[92] The threshold for a GFR-related augmentation of CVD risk is likely to be age dependent. In studies of older individuals, the risk of excess CVD events seems to appear at an eGFR less than 45 mL/min/1.73 m^2 (stage G3b CKD),[325] but this will likely be modified by the nature and severity of comorbidities present, such as dyslipidemia, diabetes, or long-standing hypertension. A decreased GFR can influence the risk of CVD, independent of traditional risk factors (e.g., obesity, smoking, dyslipidemia, diabetes, hypertension), but the magnitude of the additional risk for CVD events imposed by a reduced GFR in the range of 45 to 59 mL/min/1.73 m^2 is small in older adults. For this reason, a reduced GFR is not commonly included in CVD risk prediction scoring systems (e.g., Framingham Risk Score, American College of Cardiology/American Heart Association [ACC/AHA] pooled risk prediction model).[329]

More advanced forms of CKD can augment the CVD risk by numerous mechanisms, including endothelial cell or vascular wall injury from uremic toxins, vascular ossification,

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myocardial hypertrophy and pathologic remodeling, chronic volume expansion, chronic inflammation, uremic dyslipidemia (proatherogenic high-density lipoprotein [HDL], hypertriglyceridemia), arterial hypertension, and thrombotic microangiopathy. The fundamental phenomenon of aging can interact with these pathologic processes in numerous ways that are difficult to disentangle unambiguously.

END-STAGE KIDNEY DISEASE AND AGING

A requirement for treatment by kidney replacement modalities (kidney replacement therapy [KRT]; dialysis and/or transplantation) is not uncommon in older adults. According to the U.S. Renal Data System (USRDS) Annual Report in 2021, the age-specific annual incidence rate for RRT in 2019 for subjects 75 years of age or older was 1587 per million population (pmp) and, for those 65 to 74 years, was 1307 pmp (www.usrds.org). Both values have been slowly declining (overall about 15%, for uncertain reasons) since 2009 but are still well above the RRT incident rate of 622 pmp/year for those aged 45 to 64 years and 123 pmp for those 18 to 44 years. As has been the case for many years, men outnumber women for KRT incidence by about 20% to 30%. For non-KRT-requiring CKD, the male-to-female ratio is opposite (www.usrds.org). However, this may in part be confounded by the sex coefficient in eGFR creatinine equations, which gives women lower eGFR values, and the lower muscle mass (and creatinine generation) in women than men, which increases their UACR. An important consideration of the development of KRT-requiring ESKD is the competing risk of death, usually from a CVD or neoplasia event. This risk is magnified in older adults so that at an advanced age, any patient with CKD stage G3 (eGFR, 30–59 mL/min/1.73 m^2) is more likely to die than reach ESKD.[330]

In a landmark study, Eriksen and Ingebretsen conducted a 10-year, population-based study in Tromso, Norway, including participants (median age, 75 years; 25%, 75% range 67.7–80.4 years) with confirmed CKD stage G3.[331] About 70% of the participants experienced some decline in eGFR over time. The 10-year cumulative risk of reaching treated ESKD was 4% (higher in men than in women), whereas the cumulative risk of death was 51% (higher in men than in women). The prognosis of CKD and the eventual likelihood of requiring KRT was highly sex dependent and greatly influenced by the competing risk of death. In another key study, O'Hare and coworkers[330] studied 209,622 U.S. veterans (97% male; mean age, 73 ± 9 years) with stages G3 to G5 CKD followed for a mean of 3.2 years. Although the rates of death and ESKD were inversely related to the eGFR value at baseline irrespective of age, at comparable levels of eGFR, older age was associated with higher death rates and lower rates of KRT. The threshold of eGFR below which the rate of ESKD exceeded the risk of death was about 15 mL/min/1.73 m^2 for veterans 65 to 84 years of age. As with the Eriksen and Ingebretsen study,[331] the rate of decline of the eGFR was slower in older veterans. The effect of sex could not be formally examined in this study. Nevertheless, both studies indicated that age and sex are key factors that influence the transition from CKD (stages G3–G5) to treated ESKD.

The impact of age on the association of eGFR and albuminuria on outcomes of mortality and treated ESKD was also studied in an exhaustive analysis of 2,051,244 participants from 33 general population or high-risk CVD cohorts and 13 CKD cohorts by Hallan and colleagues[322] in the CKD Prognosis Consortium. In most of these cohorts, eGFR values were determined only once, and CKD was not confirmed by the duration criteria, leading to the possibility of overestimating the prevalence of CKD. In addition, the analysis used a single reference value of 80 mL/min/1.73 m^2 for the eGFR. The study showed a roughly comparable hazard ratio for the risk of ESKD with declining eGFR at all ages, but the average absolute rate of ESKD declined substantially with age as a function of the declining eGFR, consistent with the observations discussed earlier. The relative and absolute risks for ESKD increased as the UACR rose above 10 mg/g, but the magnitude of these risks was somewhat blunted in persons >75 years). Although a threshold of about 60 mL/min/1.73 m^2 was noted for defining an increased relative risk of ESKD at all ages, the choice of a single reference comparator of 80 mL/min/1.73 m^2 (rather than the lowest risk group for age as the reference comparator) might have influenced the findings, as discussed earlier in relation to the association of eGFR with all-cause mortality at varying ages). It is noteworthy that the absolute rate of ESKD in the very old (>75 years) did not increase above baseline values until the eGFR was well below 45 mL/min/1.73 m^2.

In a longitudinal study conducted by Shardlow and colleagues[332,333] in the United Kingdom, older participants (mean age, 73 years) with stage G3 CKD (mean eGFR, 53 mL/min/1.73 m^2) had a low prevalence of abnormal albuminuria (16%). After 5 years of additional follow-up, they had a low prevalence of ESKD (0.2%), progression to a higher stage of CKD occurred in 17.7%, remissions of CKD developed in 19.3%, and stable kidney function was observed in 34.1%. Albuminuria at diagnosis was a key risk factor for progression. In this study, 14.2% died before the 5-year follow-up and 14.8% were lost to follow-up.

ACUTE KIDNEY INJURY AND DISEASES OF THE KIDNEY AND URINARY TRACT IN AGING

A strong relationship exists between the incidence and prevalence of AKI and age.[334] The pathophysiologic mechanisms underlying this association are complex and multifactorial (discussed further in Chapter 27). The physiologic reduction of nephron number and GFR with advancing age, discussed earlier, alters the pharmacokinetics of many water-soluble drugs. This can lead to an accumulation of nephrotoxic levels of agents in body fluids, leading to AKI.[335] Dosage modifications, primarily in the interval between dosing, rather than in loading dosage, are required to minimize such risks.[195] Agents that affect glomerular hemodynamics, such as NSAIDs, RAAS inhibitors, or SGLT2 inhibitors, can cause a reduction in the GFR in older individuals that is more profound than that in younger individuals, particularly when combined with some degree of extracellular volume depletion. Older individuals also may have a comorbidity that exposes them to an episode of AKI, such as heart failure, diabetes, urosepsis, and cancer,

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and require more interventions that may predispose to AKI, such as surgery and angiography. As discussed later, aging is also associated with an increased prevalence of diseases of the kidney and urinary tract that may directly cause AKI.

DISEASES OF THE KIDNEY AND URINARY TRACT IN AGING

The occurrence of specific diseases of the kidneys and urinary tract can be influenced markedly by a person's age. These diseases are discussed in detail in other chapters and will not be elaborated further here. Certain glomerular and vascular diseases have a predilection to affect older adults (Box 21.2).[32,336-338] Benign prostatic hypertrophy is common in older men and can result in lower urinary tract obstruction and CKD, sometimes irreversible. Obstructive uropathy can also be the consequence of ureteral obstruction in gynecologic cancer in older women and bladder dysfunction associated with neurologic diseases and diabetes.

CAN RENAL AGING BE MODIFIED?

As the molecular mechanisms of organ senescence are gradually revealed, it is natural to ask about their potential for modification.[339-342] This is a difficult question because the biologic processes of aging are so complex and are active over an extended time period. In addition, it is necessary to make a clear distinction between the effects of interventions directed at diseases associated with aging (e.g., better glycemic control in diabetes, avoidance of obesity, control of hypertension) and the modification of the fundamental aging process itself. So far, no elixir of life or "fountain of youth" has been discovered, but several promising avenues have been identified. Caloric restriction, mTOR inhibition, and RAAS blockade all retard aging across many animal species,[343] but their effects in humans are not well understood. It is possible that all three of these approaches converge on mitochondrial energy production to slow aging. The angiotensin II type I receptor may play a pivotal role in renal senescence,[343] independent of angiotensin II, perhaps via a sirtuin-linked mechanism.[343,344] However, because normal healthy aging, with the steady attrition of glomeruli, is not usually accompanied by albuminuria or a rise in the snGFR (see earlier), the prospects of RAAS inhibition as a means of retarding the rate of renal aging are not promising. Caloric restriction seems to cause a reduction in renal fibrosis (via mTOR inhibition and 5′-adenosine monophosphate-activated protein kinase [AMPK] activation).[37,344] Calorie restriction and mTOR inhibition activate sirtuin 1, which has antiaging properties. Other agents, such as the biguanide metformin, may also have antiaging effects via an action on the AMPK/mTOR pathway.[345] Telomere shortening can be modulated by telomerase activity, but whether the manipulation of telomere length will be safe and effective in delaying organ senescence is unknown. Modulating Klotho expression pharmacologically might have beneficial effects via reduced oxidative stress. Many pharmacologic agents, dietary supplements, and lifestyle modification can increase serum Klotho concentrations.[346] mTOR inhibition increases serum Klotho concentrations and enhances autophagy.[347] Elimination of senescent cells (senolytic therapy) by targeting apoptosis of senescent cells (TASCs) via disruption of the FOX04 peptide-p53 interaction shows great promise in improving kidney function in aging mice.[348,349] Resveratrol, a polyphenol present in many pigmented vegetables and fruits, can ameliorate aging as a calorie restriction mimetic, possibly by improving the metabolic profile and inhibiting cyclic adenosine monophosphate (cAMP) phosphodiesterases and the AMPK pathway.[350] Because an imbalance of energy supply and demand may underlie aging, a reprogramming of metabolism from anaerobic glycolysis to aerobic glycolysis might be an effective strategy.[6] Antifibrosis regimens, such as with the transforming growth factor beta (TGF-β)/smad pathway inhibition, BMP-7 administration, or the use of other agents, represent a promising avenue of research,[351,352] especially in the domain of renal aging. Translating studies carried out in lower animals (e.g., *Caenorhabditis elegans*, mice, rats) to the human organism will be tedious, time consuming, and fraught with obstacles. Armed with new tools, however, such as nephron number quantification, it may be possible to test some of the most promising strategies directly in humans. Hopefully, in the not-too-distant future, renal aging will not be inevitable, even though immortality will continue to be an elusive goal.

Box 21.2 Kidney and urinary tract diseases that occur more commonly in older adults

Glomerular and vascular diseases

Crescentic glomerulonephritis due to systemic or renal-limited antineutrophil cytoplasmic autoantibody (ANCA)–associated vasculitis
Membranous nephropathy (primary and cancer related)
Monoclonal immunoglobulin deposition diseases (amyloid light-chain amyloidosis and nonamyloid types)
Diabetic nephropathy
Fibrillary glomerulonephritis
Anti–glomerular basement membrane (GBM) disease (mainly females)
Atheroembolic disease
Atheromatous renovascular stenosis

Tubulointerstitial diseases

Acute kidney injury (toxic or ischemic cause)
Lymphomatous infiltration
Renal cell cancer

Urinary tract disorders

Lower urinary tract obstruction due to prostate disease
Ureteric obstruction from cancer (cervical cancer in females)
Transitional cell carcinoma of the bladder
Prostate cancer

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QUESTIONS

1. You have been asked to see a man 78 years of age who has an estimated glomerular filtration rate (eGFR) of 45 mL/min/1.73 m^2 with a question about the need to start an angiotensin-converting enzyme (ACE) inhibitor. He has essentially been healthy apart from having two inguinal hernia repairs, and he has mild symptoms of prostatism with nocturia twice a night. Otherwise, he feels well. Electrolyte levels are within normal limits, with a potassium level of 5.0 mmol/L. His urinary albumin-to-creatinine ratio is 12 mg/g of creatinine. Which outcomes are most likely in this patient?
   a. He will progress to end-stage kidney disease (ESKD) within the next 5 years.
   b. He will progress to ESKD within the next 10 years.
   c. He will die before the development of ESKD.
   d. Kidney function will remain stable.
   e. Kidney function will improve.

Answer: c and d

Rationale: In the absence of albuminuria, it is unlikely this patient has significant kidney disease. The GFR is low, most likely because of the loss of glomeruli with age. An ACE inhibitor in this patient is not indicated for his kidney function.

2. A 72-year-old woman is admitted after a fall. She was found 24 hours later at home. It is noted that her serum sodium level is 156 mmol/L. Her estimated GFR is 48 mL/min/1.73 m^2. She is on no medication and lives alone. Which age-related factors may have contributed to the hypernatremia?
   a. Reduced thirst
   b. Reduced urinary concentrating capacity
   c. Reduced total body water
   d. Increased vasopressin secretion
   e. High salt intake

Answer: a, b, c

Rationale: All these factors predispose to hypernatremia in older adults. Older kidneys have a reduced capacity to conserve water and solutes. Total body water is reduced (replaced with greater fat mass), and therefore older adults are more prone to volume depletion. Older adults have a reduced thirst response to increased osmolality and therefore are less likely to be able to correct hypernatremia spontaneously.

3. A 76-year-old woman wants to donate a kidney to her 46-year-old son and is coming to you for a second opinion, given that she was declined as a donor elsewhere because her GFR is below 60 mL/min/1.73 m^2. Her body mass index (BMI) is 21. Her eGFR, calculated by the CKD-EPI formula, is 57 mL/min/1.73 m^2. She has no hypertension or diabetes, is a nonsmoker, and is a vegetarian. She exercises regularly, taking long walks three times a week with a seniors group. She is adamant that she wants to donate a kidney to her son—she sees him suffering on dialysis and wants to do everything possible to help him. Which of the following may assist in assessment of this woman as a potential donor?
   a. Determination of a cystatin C GFR
   b. Measurement of her GFR before and after a protein load (renal functional reserve)
   c. Determination of 24-hour urinary albumin excretion
   d. A renal biopsy showing the presence of 10% focal and segmental glomerulosclerosis
   e. Determination of kidney size by ultrasound

Answer: All the above

Rationale: Given her relatively low BMI, use of a cystatin C GFR may assist in determining the accuracy of the creatinine-based eGFR value. Because she is a vegetarian, the GFR may be lower than expected. Although not routinely used, detection of the capacity of her kidneys to respond to a protein load may indicate preservation of renal reserve. Any evidence of albuminuria or the presence of focal segmental glomerulosclerosis (FSGS) on the biopsy would indicate a primary renal pathology and likely exclude the woman from donation. Similarly, significant renal asymmetry or small kidneys would indicate potential renal ischemia or renal disease and would be exclusion factors for donation.

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